Fifth Third Bank - Watson

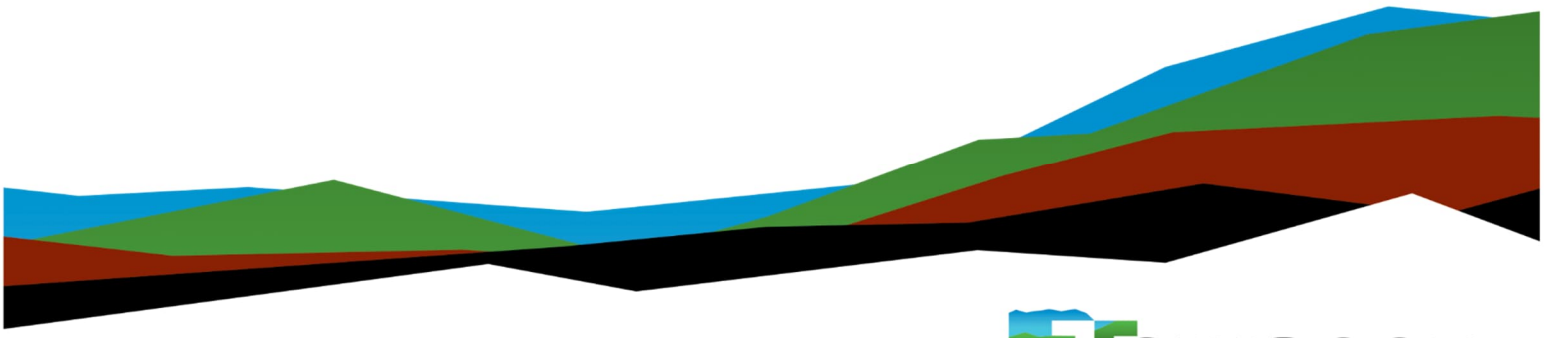
Geotechnical Engineering Report

Warner Robins, Georgia

May 4, 2026 | Terracon Project No. 49265099

Prepared for:

Praxis3
100 Peachtree Street NW, Suite 1450
Atlanta, GA 30303



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May 4, 2026

Praxis3
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Attn: Ms. Nicole Webb
P: 678-904-7452
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Re: Geotechnical Engineering Report
Fifth Third Bank - Watson
2711 Watson Boulevard
Warner Robins, Georgia
Terracon Project No. 49265099

Dear Ms. Webb:

We have completed the scope of Geotechnical Engineering services for the above referenced project in general accordance with Terracon Proposal No. P49265099 dated April 3, 2026. This report presents the findings of the subsurface exploration and provides geotechnical recommendations concerning earthwork and the design and construction of foundations, floor slabs, and pavements for the proposed project.

We appreciate the opportunity to be of service to you on this project. If you have any questions concerning this report or if we may be of further service, please contact us.

Sincerely,

Terracon



Ethan Fraser, GIT
Project Engineer

Daimia T. Gunning-Gayle, P.E.
Senior Engineer

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Supporting Information

Note: This report was originally delivered in a web-based format. Blue Bold text in the report indicates a referenced section heading. The PDF version also includes hyperlinks which direct the reader to that section and clicking on the Terracon logo will bring you back to this page. For more interactive features, please view your project online at client.terracon.com.

Refer to each individual Attachment for a listing of contents.

Report Summary

This summary should be used in conjunction with the entire report for design purposes. It should be recognized that details were not included or fully developed in this section, and the report must be read in its entirety for a understanding of the items contained herein.

Report Summary

Topic [1]	Overview Statement [2]
Project Description	About 1,900 square feet single-story structure Assumed maximum loads: columns: 200 kips, wall loads: 4 kips/ft, slabs: 100 to 150 psf
Geotechnical Characterization	■ Some areas of existing fill to about 3 feet deep. ■ Native soils, coastal plain soils, were encountered in each boring. ■ Groundwater was not encountered while drilling in any of borings.
Earthwork	Earthwork is anticipated to include excavations and engineered fill placement. Remove existing fill where observed or Geotechnical engineer to further evaluate areas where existing fill is present after site stripping.
Shallow Foundations	Shallow foundations are recommended for building support Allowable bearing pressure = 3,000 psf – foundations bearing upon native soils, existing fill materials, or newly placed structural fill Expected settlements: less than 1-inch total, about ½ inch differential.
Pavements	With subgrade prepared as noted in Earthwork. Both rigid (concrete) and flexible (asphalt) pavement sections will be considered. In the absence of specific traffic loading data, typical pavement design parameters will be considered.
General Comments	This section contains important information about the limitations of this geotechnical engineering report.

1. If the reader is reviewing this report as a pdf, the topics above can be used to access the appropriate section of the report by simply clicking on the topic itself.
2. This summary is for convenience only. It should be used in conjunction with the entire report for design purposes.

Introduction

This report presents the results of our subsurface exploration and Geotechnical Engineering services performed for the proposed Fifth Third Bank to be located at 2711 Watson Boulevard in Warner Robins, Georgia. The purpose of these services was to provide information and geotechnical engineering recommendations relative to:

- Subsurface soil conditions
- Groundwater conditions
- Seismic site classification per IBC
- Site preparation and earthwork
- Demolition considerations
- Foundation design and construction
- Floor slab design and construction
- Lateral earth pressure
- Pavement design and construction

The geotechnical engineering Scope of Services for this project included the advancement of test borings, laboratory testing, engineering analysis, and preparation of this report.

Drawings showing the site and boring locations are shown on the Site Location and Exploration Plan, respectively. The results of the laboratory testing performed on soil samples obtained from the site during our field exploration are included on the boring logs and as separate graphs in the Exploration Results section.

Project Description

Our initial understanding of the project was provided in our proposal and was discussed during project planning. A period of collaboration has transpired since the project was initiated, and our final understanding of the project conditions is as follows:

Project Description

Item	Description
Information Provided	The following files were provided: ■ 2026-02-05_Fifth Third Watson_TFF 1B_approved.pdf ■ Fifth Third Watson_boring locations.pdf
Project Description	The project consists of a new bank and associated access drives/parking.

Project Description

Item	Description
Proposed Structure	Structures associated with the project include the development of a new Fifth Third Bank building, parking lot and drives, a trash enclosure, and two banking lanes.
Building Construction	The bank building will be approximately 1,900 square feet. Drives and parking surround the bank on all sides. Two banking lanes are located directly south of the bank. The trash enclosure is located in the southeastern corner of the proposed development.
Finished Floor Elevation	Not provided, but assumed to be close to existing grades.
Maximum Loads	Anticipated structural loads were not provided. In the absence of information provided by the design team, we used the following loads in estimating settlement based on our experience with similar projects. ■ Columns: up to 200 kips ■ Walls: up to 4 kips per linear foot (klf) ■ Slabs: 100 to 150 pounds per square foot (psf)
Grading/Slopes	Proposed finished grade elevation for the building pad is expected to be at or near existing grades.
Below-Grade Structures	None planned (basement, underground detention, etc.).
Free-Standing Retaining Walls	None planned, but may become necessary as design progresses.
Pavements	A preferred pavement surfacing has not been identified to us as part of the preliminary information. Asphalt/concrete surfacing is the assumed preference. Traffic information has not been provided.
Building Code	2024 IBC.

Terracon should be notified if any of the above information is inconsistent with the planned construction, especially the grading limits, as modifications to our recommendations may be necessary.

Site Conditions

The following description of site conditions is derived from our site visit in association with the field exploration and our review of publicly available geologic and topographic maps.

Site Conditions

Item	Description
Parcel Information	The project is located at 2711 Watson Boulevard in Warner Robins, Georgia. Latitude/Longitude (approximate): 32.6178° N / 83.6714° W See Site Location
Existing Improvements	The site is currently comprised of two commercial buildings and asphalt-paved parking/drives.
Current Ground Cover	Asphalt paved with some landscape areas around the existing buildings.
Existing Topography	The site is relatively flat.

Geotechnical Characterization

Site Geology

Based on a review of the Geologic Map of Georgia, dated 1976, as published by the Georgia Geological Survey, the subject site is located within the Coastal Plain Physiographic Province of Georgia. The Coastal Plain Province is a wedge-shaped deposit of Cretaceous and younger sediments which range in thickness from near zero at the contact with the Piedmont Physiographic Province (Fall Line) along its northwest edge, to thousands of feet at the coast. Coastal Plain soils are marine deposits which range in age from as old as 130 million years near the “Fall Line” contact with ancient continental rocks to recent deposits near the coast. They contain various materials including interbedded gravels, sands, silts and clays, soft to hard limestone and organic materials such as peat.

Typical Subsurface Profile

The borings drilled at the site generally encountered surficial materials, existing fill materials, and native soils.

Fill soils are those soils that have been placed or reworked by man in conjunction with past construction grading, underground utility installation, farming or other previous activity at the site. Fill can be composed of different soil types from various sources and can contain debris from building demolition, organics, topsoil, trash, etc. The engineering properties of the fill depend primarily on its composition, density, and moisture content.

Five of the eight soil test borings (borings B-1 and P-1 through P-4) performed encountered existing fill to a depth of about 3 feet below existing grades. The fill appears to have been placed in a controlled manner, but we have no records to indicate the degree of compaction achieved during placement. If records are available, they should be provided to Terracon for use in our evaluation and development of geotechnical recommendations.

Based on the results of the borings, subsurface conditions on the project site can be generalized as follows:

Typical Subsurface Profile

Description	Approximate Depth to Bottom of Stratum (Feet)	Material Encountered	Consistency/Density
Stratum 1	2 to 3 inches 0 to 12 inches	Asphalt Aggregate Base Course	--
Stratum 2 [1]	3 feet	*Fill* – Sandy Lean Clay (CL).	Firm to Very Stiff
Stratum 3	Beyond boring termination depths	*Native/Coastal Plain Soils* – Sandy Lean Clay (CL) and Clayey Sands (SC).	Stiff to Very Stiff and Medium Dense to Dense

1. Encountered in borings B-1 and P-1 through P-4.

Specific conditions encountered at each boring location are indicated on the individual boring logs. Stratification boundaries on the boring logs represent the approximate location of changes in soil types; in situ, the transition between materials may be gradual. Details for each of the borings are presented on the boring logs included in the Exploration Results.

GeoModel

We have developed a general characterization of the subsurface conditions based upon our review of the subsurface exploration, laboratory data, geologic setting and our understanding of the project. This characterization, termed GeoModel, forms the basis of our geotechnical calculations and evaluation of the site. Conditions observed at each exploration point are indicated on the individual logs. The GeoModel can be found in the Exploration Results attachment of this report.

As part of our analyses, we identified the following model layers within the subsurface profile. For a more detailed view of the model layer depths at each boring location, refer to the GeoModel.

GeoModel

Model Layer	Layer Name	General Description
1	Surficial Materials	Asphalt, Aggregate Base Course
2	Fill	Fill – Sandy Lean Clay
3	Native / Coastal Plain Soils	Native/Coastal Plain Soils – Sandy Lean Clay and Clayey Sand

Laboratory Testing

Laboratory tests were conducted on selected soil samples and the test results are presented in the Exploration Results, and are summarized in the following tables.

Natural Moisture Content

Boring Number	Depth (Feet)	Natural Moisture Content (%)
B-1	3.5 – 5	20.5
B-3	1 – 2.5	20.6
P-3	3.5 – 5	20.6

Atterberg Limits

Boring Number	Depth (Feet)	Liquid Limit	Plastic Limit	Plasticity Index
B-1	3.5 – 5	37	19	18
B-3	1 – 2.5	41	18	23
P-3	3.5 – 5	38	18	20

Grain Size Analysis

Boring Number	Depth (Feet)	Gravel (%)	Sand (%)	Fines (%)
B-1	3.5 – 5	0	47.0	53.0
B-3	1 – 2.5	0.9	39.1	60
P-3	3.5 – 5	0.3	48.4	51.3

Groundwater

The boreholes were observed while drilling and after completion for the presence and level of groundwater. Groundwater was not observed in the borings while drilling, or for the short duration that the borings were allowed to remain open. However, this does not necessarily mean the borings terminated above groundwater. Due to the low permeability of the soils encountered in the borings, a relatively long period of time may be necessary for a groundwater level to develop and stabilize in a borehole in these materials. Long term observations in piezometers or observation wells sealed from the influence of surface water are often required to define groundwater levels in materials of this type.

Groundwater level fluctuations occur due to seasonal variations in the amount of rainfall, runoff and other factors not evident at the time the borings were performed. In addition, perched water can develop over low permeability soil or rock strata. Therefore, groundwater levels during construction or at other times in the life of the structure may be higher or lower than the levels indicated on the boring logs. The possibility of groundwater level fluctuations should be considered when developing the design and construction plans for the project.

Seismic Site Class

The seismic design requirements for buildings and other structures are based on Seismic Design Category. Site Classification is required to determine the Seismic Design Category for a structure. Per the 2024 International Building Code (2024 IBC), the seismic site classification is based on a site subsurface profile extending to a depth of 100 feet. Based on the soil conditions observed at the site and as described on the exploration logs and results, it is our professional opinion that a default Seismic Site Classification of D be considered for the project. Subsurface explorations at this site were only extended to a maximum depth of 20 feet. Therefore, the recommended seismic site class was selected to represent the most critical site conditions per ASCE 7-22 Section 20.1 as referenced by the 2024 IBC. Additional geophysical testing may be performed to confirm or potentially improve the site class.

Geotechnical Overview

The site appears suitable for the proposed construction based upon geotechnical conditions encountered in the test borings, provided that the recommendations presented in this report are implemented in the design and construction phases of this project.

If the site has been prepared in accordance with the requirements noted in the Earthwork section, the proposed structure can be supported by a shallow, spread footing foundation system bearing on native/coastal plain soils or newly placed structural fill extending to native soil. Additional site preparation recommendations, including subgrade improvement and fill placement, are provided in the Earthwork section.

The Shallow Foundations section addresses support of the building bearing on native soil, existing fill, or newly placed structural fill extending to native soil. The Floor Slabs section addresses slab-on-grade support of the building.

We anticipate that an asphaltic concrete pavement system will generally be considered for this site. The Pavements section addresses the design of pavement systems.

Five of the eight soil test borings (borings B-1 and P-1 through P-4) performed encountered existing fill to a depth of about 3 feet below existing grades. Support of floor slabs, foundations, and pavements on or above existing fill materials is discussed in this report. However, even with the recommended construction procedures, an inherent risk remains for the owner that compressible fill or unsuitable material, within or buried by the fill, will not be discovered. This risk of unforeseen conditions cannot be eliminated without completely removing the existing fill but can be reduced by following the recommendations contained in this report. To take advantage of the cost benefit of not removing the entire amount of undocumented fill, the owner must be willing to accept

the risk of increased differential performance which can result in increased cracking and abrupt differential settlement. Should this risk be acceptable, floor slabs, foundations, and pavements can be supported above the fill.

The recommendations contained in this report are based upon the results of field and laboratory testing (presented in the Exploration Results), engineering analyses, and our current understanding of the proposed project. The General Comments section provides an understanding of the report limitations.

Earthwork

Earthwork is anticipated to include demolition, clearing and grubbing of landscaped areas, excavations, and engineered fill placement. The following sections provide recommendations for use in the preparation of specifications for the work. Recommendations include critical quality criteria, as necessary, to render the site in the state considered in our geotechnical engineering evaluation for foundations, floor slabs, and pavements.

Demolition

The proposed building will be constructed within the footprint of an existing commercial development which will need to be demolished.

We recommend that foundations, slabs, or utilities be removed from within the proposed building footprint and at least 5 feet beyond the outer edge of foundations.

For areas outside the proposed building footprints and foundation bearing zones, existing pavement, foundations, slabs, or utilities (if encountered) should be removed where they conflict with proposed utilities, retaining walls, and pavements. In such cases, existing foundations, floor slabs, and utilities should be removed to a depth of at least 2 feet below the affected utility or design pavement subgrade elevation.

Site Preparation

Prior to placing fill, existing vegetation, topsoil, and root mats should be removed. Complete stripping of the topsoil should be performed in the proposed building and parking/driveway areas.

Evidence of underground facilities (such as utilities) was observed during the exploration and site reconnaissance and such features could be encountered during construction. If unexpected fills or underground facilities are encountered, such features should be

removed, and the excavation thoroughly cleaned prior to backfill placement and/or construction.

Subgrade Preparation

After stripping, proofrolling should be performed with a fully loaded tandem-axle dump truck. A Terracon geotechnical engineer or their representative should observe proofrolling to aid in locating unstable subgrade materials. Proofrolling should be performed after a suitable period of dry weather to avoid degrading an otherwise acceptable subgrade and to reduce the amount of undercutting / remedial work required. Unstable materials identified should be stabilized as directed by the engineer based on conditions observed during construction. Undercut and replacement and densification in-place are typical remediation methods.

All exposed areas which will receive fill, should be scarified to a minimum depth of 10 inches, moisture conditioned as necessary, and compacted per the compaction requirements in this report. Compacted structural fill soils should then be placed to the proposed design grade and the moisture content and compaction of subgrade soils should be maintained until foundation or pavement construction.

Based upon the subsurface conditions determined from the geotechnical exploration, subgrade soils exposed during construction are anticipated to be relatively workable; however, the workability of the subgrade may be affected by precipitation, repetitive construction traffic or other factors. If unworkable conditions develop, workability may be improved by scarifying and drying.

Existing Fill

As noted in Geotechnical Characterization, five of the eight soil test borings (borings B-1 and P-1 through P-4) performed encountered existing fill to a depth of about 3 feet below existing grades. The fill appears to have been placed in a controlled manner, but we have no records to indicate the degree of compaction achieved during placement. If records are available, they should be provided to Terracon for use in our evaluation and development of geotechnical recommendations.

Support of footings, floor slabs, and pavements on or above existing fill soils is discussed in this report. However, even with the recommended construction procedures, inherent risk exists for the owner that compressible fill or unsuitable material, within or buried by the fill will, not be discovered. This risk of unforeseen conditions cannot be eliminated without completely removing the existing fill but can be reduced by following the recommendations contained in this report.

If the owner elects to construct footings and floor slabs on the existing fill to reduce initial construction costs in exchange for increased potential longer-term distress, the following protocol should be followed. After the planned grading has been completed, the entire area should be proofrolled with heavy, rubber tire construction equipment, to aid in delineating areas of soft or otherwise unsuitable soil. Areas of soft or otherwise unsuitable material should be undercut and replaced with either new structural fill or suitable, existing on-site materials. The existing undocumented fill that was removed can be evaluated for reuse as structural fill.

If the owner elects to construct pavements on the existing fill, the following protocol should be followed. Once the planned subgrade elevation has been reached, the entire pavement area should be proofrolled. Areas of soft or otherwise unsuitable material should be undercut and replaced with either new structural fill or suitable, existing on site materials.

Excavation

After comparison of the boring data and the assumed finished grade elevations, we do not anticipate that materials requiring difficult excavation techniques for removal will be encountered during mass grading operations or utility trench installation. The existing fill and native/coastal plain soils found above assumed finished floor levels should be excavatable with conventional earthmoving equipment.

Fill Material Types

Fill required to achieve design grade should be classified as structural fill and general fill. Structural fill is material used below, or within 10 feet of structures, pavements or constructed slopes. General fill is material used to achieve grade outside of these areas.

Reuse of On-Site Existing Fill Soil: A large portion of the existing fill is expected to be suitable for reuse as new fill provided it is free of organics, debris, and unsuitable materials. Terracon should evaluate existing fill materials for use.

Soils used for structural and general fill should meet the following material property requirements:

Fill Material Types

Soil Type[1,2,3]	USCS Classification	Acceptable Parameters (for Structural Fill)
Fine Grain	CL and ML	Liquid Limit less than 45 Plasticity index less than 25

Fill Material Types

Soil Type[1,2,3]	USCS Classification	Acceptable Parameters (for Structural Fill)
Granular	SP, SM, SC, and SW	Less than 50% passing No. 200 sieve
On-Site Soils [4]	CL and SC	Liquid Limit less than 45 Plasticity index less than 25

1. Structural and general fill should consist of approved materials free of organic matter and debris. Frozen material should not be used, and fill should not be placed on a frozen subgrade. A sample of each material type should be submitted to the Geotechnical Engineer for evaluation prior to use on this site.
2. All fill should have a maximum dry density of at least 95 pounds per cubic foot (pcf) as determined by the standard Proctor test (ASTM D 698).
3. Any materials proposed as fill from off-site sources should be tested for compliance with these criteria before being hauled to the site.
4. Re-use of existing fill should meet the requirements noted above.

Fill Placement and Compaction Requirements

Structural and general fill should meet the following compaction requirements:

Fill Placement and Compaction Requirements

Item[,2,3]	Structural Fill	General Fill
Maximum Lift Thickness	8 inches or less in loose thickness when heavy, self-propelled compaction equipment is used 4 to 6 inches in loose thickness when hand-guided equipment (i.e. jumping jack or plate compactor) is used	Same as structural fill
Minimum Compaction Requirements[1,2]	98% of max. dry density within 1 foot of floor slabs or finished pavement subgrades 95% of max. dry density below foundations and more than 1 foot below finished floor slabs or pavement subgrades	92% of max.
Water Content Range[1,2,3]	Fine Grain Soils: -2% to +3% of optimum Granular: -2% to +3% of optimum	As required to achieve min. compaction requirements

Fill Placement and Compaction Requirements

Item[1,2,3]	Structural Fill	General Fill

1. Maximum density and optimum water content as determined by the standard Proctor test (ASTM D 698).
2. Fill should be tested for compaction and moisture content during placement. Should the results of the in-place density tests indicate that the specified moisture or compaction requirements have not been met, the area represented by the test should be reworked and retested as required until the specified moisture and compaction requirements are achieved.
3. Moisture levels should be maintained low enough to allow for satisfactory compaction to be achieved without pumping when proofrolled.

Utility Trench Backfill

Any soft or unsuitable materials encountered at the bottom of utility trench excavations should be removed and replaced with structural fill or bedding material in accordance with public works specifications for the utility being supported. This recommendation is particularly applicable to utility work requiring grade control and/or in areas where subsequent grade raising could cause settlement in the subgrade supporting the utility. Trench excavation should not be conducted below a downward 1:1 projection from existing foundations without engineering review of shoring requirements and geotechnical observation during construction.

On-site materials are considered suitable for backfill of utility and pipe trenches from 1 foot above the top of the pipe to the final ground surface, provided the material is free of organic matter and deleterious substances.

Trench backfill should be mechanically placed and compacted as discussed earlier in this report. Compaction of initial lifts should be accomplished with hand-operated tampers or other lightweight compactors.

Grading and Drainage

Adequate positive drainage should be provided during construction and maintained throughout the life of the development to prevent an increase in moisture content of the foundation, pavement and backfill materials. Surface water drainage should be controlled to prevent undermining of fill slopes and structures during and after construction.

Gutters and downspouts that drain water a minimum of 10 feet beyond the footprint of the proposed structures are recommended. This can be accomplished through the use of

splash-blocks, downspout extensions, and flexible pipes that are designed to attach to the end of the downspout. Flexible pipe should only be used if it is daylighted in such a manner that it gravity-drains collected water. Splash-blocks should also be considered below hose bibs and water spigots.

It is recommended that all exposed earth slopes be seeded to provide protection against erosion as soon as possible after completion. Seeded slopes should be protected until the vegetation is established. Sprinkler systems should not be installed behind or in front of walls or near slopes.

Groundwater Considerations

Groundwater was not encountered in any of the borings during our exploration. Based on our understanding of the proposed development, we do not expect groundwater to affect construction. If groundwater is encountered during construction, some form of temporary or permanent dewatering may be required. Conventional dewatering methods, such as pumping from sumps, should likely be adequate for temporary removal of any groundwater encountered during excavation at the site. Well points would likely be required for significant groundwater flow, or where excavations penetrate groundwater.

Earthwork Construction Considerations

Upon completion of filling and grading, care should be taken to maintain the subgrade water content prior to construction of grade-supported improvements such as floor slabs and pavements. Construction traffic over the completed subgrades should be avoided. The site should also be graded to prevent ponding of surface water on the prepared subgrades or in excavations. Water collecting over or adjacent to construction areas should be removed. If the subgrade freezes, desiccates, saturates, or is disturbed, the affected material should be removed, or the materials should be scarified, moisture conditioned, and recompacted prior to floor slab construction.

As a minimum, excavations should be performed in accordance with OSHA 29 CFR, Part 1926, Subpart P, “Excavations” and its appendices, and in accordance with any applicable local and/or state regulations.

Construction site safety is the sole responsibility of the contractor who controls the means, methods, and sequencing of construction operations. Under no circumstances shall the information provided herein be interpreted to mean Terracon is assuming responsibility for construction site safety or the contractor's activities; such responsibility shall neither be implied nor inferred.

Excavations or other activities resulting in ground disturbance have the potential to affect adjoining properties and structures. Our scope of services does not include review of available final grading information or consider potential temporary grading performed by the contractor for potential effects such as ground movement beyond the project limits. A preconstruction/ precondition survey should be conducted to document nearby property/infrastructure prior to any site development activity. Excavation or ground disturbance activities adjacent or near property lines should be monitored or instrumented for potential ground movements that could negatively affect adjoining property and/or structures.

Construction Observation and Testing

The earthwork efforts should be observed by the Geotechnical Engineer (or others under their direction). Observation should include documentation of adequate removal of surficial materials (vegetation, topsoil, and pavements), evaluation and remediation of existing fill materials, as well as proofrolling and mitigation of unsuitable areas delineated by the proofroll.

Each lift of compacted fill should be tested, evaluated, and reworked, as necessary, as recommended by the Geotechnical Engineer prior to placement of additional lifts. Any areas that do not meet the compaction specifications should be reworked to achieve compliance.

In areas of foundation excavations, the bearing subgrade should be evaluated by the Geotechnical Engineer. If unanticipated conditions are observed, the Geotechnical Engineer should prescribe mitigation options.

In addition to the documentation of the essential parameters necessary for construction, the continuation of the Geotechnical Engineer into the construction phase of the project provides the continuity to maintain the Geotechnical Engineer's evaluation of subsurface conditions, including assessing variations and associated design changes.

Shallow Foundations

If the site has been prepared in accordance with the requirements noted in Earthwork, the proposed structure can be supported by a shallow, spread footing foundation system bearing on native soils or structural fill extending to native soils.

Shallow foundations may be supported on existing fill soils provided the site has been prepared in accordance with the requirements noted in the Earthwork section. However, even with the recommended construction procedures, there is inherent risk for the owner that compressible fill or unsuitable material, within or buried by the fill, will not be discovered. This risk of unforeseen conditions cannot be eliminated without

completely removing the existing fill but can be reduced by following the recommendations contained in this report. Design recommendations for shallow foundations for the proposed structure are presented in the following paragraphs.

Design Parameters – Compressive Loads

Shallow Foundation Design Parameters – Compressive Loads

Item	Description
Maximum Net Allowable Bearing Pressure[1,2]	3,000 psf
Required Bearing Stratum[3]	Native/coastal plain soils, existing fill materials, or newly placed structural fill extending to native soils
Minimum Foundation Dimensions	Columns: 24 inches Continuous: 16 inches
Ultimate Passive Resistance[4] (equivalent fluid pressures)	160 pcf (granular backfill)
Ultimate Coefficient of Sliding Resistance[5]	0.35 – granular material
Minimum Embedment below Finished Grade[6]	Exterior footings: 18 inches Interior footings in heated areas: 12 inches
Estimated Total Settlement from Structural Loads[2]	Less than about 1 inch
Estimated Differential Settlement[2,7]	About ½ of total settlement

1. The maximum net allowable bearing pressure is the pressure in excess of the minimum surrounding overburden pressure at the footing base elevation. Values assume that exterior grades are no steeper than 20% within 10 feet of structure.
2. Values provided are for maximum loads noted in Project Description. Additional geotechnical consultation will be necessary if higher loads are anticipated.
3. Unsuitable or soft soils should be over-excavated and replaced per the recommendations presented in Earthwork.
4. Use of passive earth pressures require the sides of the excavation for the spread footing foundation to be nearly vertical and the concrete placed neat against these vertical faces or that the footing forms be removed and compacted structural fill be placed against the vertical footing face. Assumes no

Shallow Foundation Design Parameters – Compressive Loads

Item	Description
	hydrostatic pressure. For passive earth pressure to develop, the footing must move horizontally to mobilize resistance. Therefore, the total calculated passive pressure has been reduced by at least one-half for design purposes. 5. Can be used to compute sliding resistance where foundations are placed on suitable soil/materials. Frictional resistance for granular materials is dependent on the bearing pressure which may vary due to load combinations. Should be neglected for foundations subject to net uplift conditions. 6. Embedment necessary to minimize the effects of frost and/or seasonal water content variations. For sloping ground, maintain depth below the lowest adjacent exterior grade within 5 horizontal feet of the structure. 7. Differential settlements are noted for equivalent-loaded foundations and bearing elevation as measured over a span of 40 feet.

The allowable foundation bearing pressures apply to dead loads plus design live load conditions. The weight of the foundation concrete below grade may be neglected in dead load computations. Interior footings should bear a minimum of 12 inches below finished grade. Finished grade is the lowest adjacent grade for perimeter footings and floor level for interior footings.

Footings, foundations, and masonry walls should be reinforced as necessary to reduce the potential for distress caused by differential foundation movement. The use of joints at openings or other discontinuities in masonry walls is recommended.

Foundation excavations should be observed by the geotechnical engineer. If the soil conditions encountered differ from those presented in this report, supplemental recommendations will be required.

Design Parameters – Overturning and Uplift Loads

Shallow foundations subjected to overturning loads should be proportioned such that the resultant eccentricity is maintained in the center-third of the foundation (e.g., $e < b/6$, where b is the foundation width). This requirement is intended to keep the entire foundation area in compression during the extreme lateral/overturning load event. Foundation oversizing may be required to satisfy this condition.

Uplift resistance of spread footings can be developed from the effective weight of the footing and the overlying soils with consideration to the IBC basic load combinations.

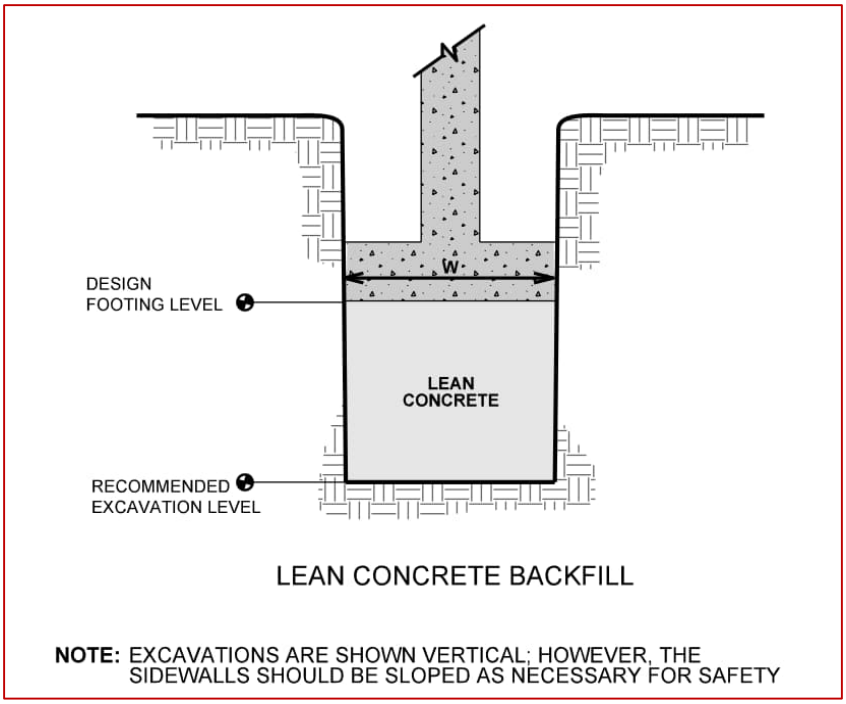
Item	Description
Soil Moist Unit Weight	120 pcf
Soil Effective Unit Weight[1]	60 pcf
Soil weight included in uplift resistance	Soil included within the prism extending up from the top perimeter of the footing at an angle of 20 degrees from vertical to ground surface

1. Effective (or buoyant) unit weight should be used for soil above the foundation level and below a water level. The high groundwater level should be used in uplift design as applicable.

Foundation Construction Considerations

As noted in Earthwork, the footing excavations should be evaluated under the observation of the Geotechnical Engineer. The base of all foundation excavations should be free of water and loose soil, prior to placing concrete. Concrete should be placed soon after excavating to reduce bearing soil disturbance. Care should be taken to prevent wetting or drying of the bearing materials during construction. Excessively wet or dry material or any loose/disturbed material in the bottom of the footing excavations should be removed/reconditioned before foundation concrete is placed.

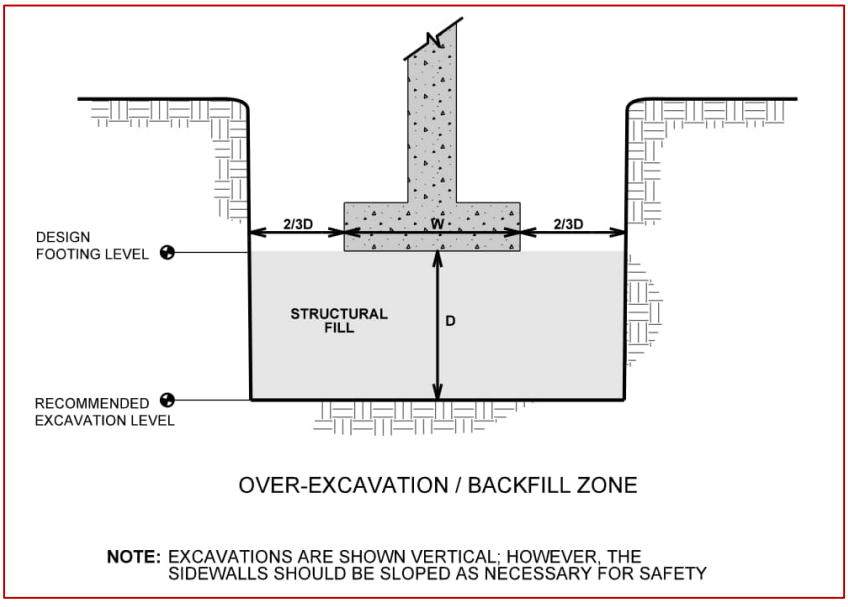
If unsuitable bearing soils are observed at the base of the planned footing excavation, the excavation should be extended deeper to suitable soils, and the footings could bear directly on these soils at the lower level or on lean concrete backfill placed in the excavations. The lean concrete replacement zone is illustrated on the sketch below.



LEAN CONCRETE BACKFILL

NOTE: EXCAVATIONS ARE SHOWN VERTICAL; HOWEVER, THE SIDEWALLS SHOULD BE SLOPED AS NECESSARY FOR SAFETY

Over-excavation for structural fill placement below footings should be conducted as shown below. The over-excavation should be backfilled up to the footing base elevation, with well-graded granular material placed as recommended in the Earthwork section.



OVER-EXCAVATION / BACKFILL ZONE

NOTE: EXCAVATIONS ARE SHOWN VERTICAL; HOWEVER, THE SIDEWALLS SHOULD BE SLOPED AS NECESSARY FOR SAFETY

For foundations adjacent to slopes, a minimum horizontal setback of 10 feet should be maintained between the foundation base and slope face. In addition, the setback should be at a location where an imaginary line extending downward at 45 degrees from the nearest edge of the foundation does not intersect the slope face.

Floor Slabs

Design parameters for floor slabs assume the requirements for Earthwork have been followed. Specific attention should be given to positive drainage away from the structure and positive drainage of the aggregate base beneath the floor slab.

In select areas, existing fill materials were observed at the site to a depth of about 3 feet below existing grade. As previously described, any existing fill present beneath floor slabs should be completely removed OR further evaluated by the Geotechnical Engineer.

Depending upon the finished floor elevation, firm soils may be observed at the floor slab subgrade level. These soils should be replaced with structural fill so the floor slab is supported on at least 2 feet of compacted suitable natural soils or structural fill.

Floor Slab Design Parameters

Floor Slab Design Parameters

Item	Description
Floor Slab Support [1]	Minimum 12 inches of approved on-site or imported soils placed and compacted in accordance with the Subgrade compacted to recommendations in Earthwork section of this report. [2]
Subbase [3]	4-inch compacted layer of free draining, granular subbase material
Estimated Modulus of Subgrade Reaction [4]	100 pounds per square inch per inch (psi/in) for point loads

1. Floor slabs should be structurally independent of building footings or walls to reduce the possibility of floor slab cracking caused by differential movements between the slab and foundation.
2. We recommend subgrades be maintained at the proper moisture condition until floor slabs are constructed. If the subgrade should become desiccated prior to construction, the affected material should be removed or the materials scarified, moistened, and recompacted. Upon completion of grading operations in the building areas, care should be taken to maintain the recommended subgrade moisture content and density prior to construction of the building floor slabs.
3. The floor slab design should include a capillary break, comprised of free-draining, compacted, granular material, at least 4 inches thick.

Floor Slab Design Parameters

Item	Description

4. Modulus of subgrade reaction is an estimated value based upon our experience with the subgrade condition, the requirements noted in Earthwork, and the floor slab support as noted in this table. It is provided for point loads. For large area loads the modulus of subgrade reaction would be lower.

The use of a vapor retarder should be considered beneath concrete slabs on grade covered with wood, tile, carpet, or other moisture sensitive or impervious coverings, when the project includes humidity-controlled areas, or when the slab will support equipment sensitive to moisture. When conditions warrant the use of a vapor retarder, the slab designer should refer to ACI 302 and/or ACI 360 for procedures and cautions regarding the use and placement of a vapor retarder.

Saw-cut contraction joints should be placed in the slab to help control the location and extent of cracking. For additional recommendations, refer to the ACI Design Manual. Joints or cracks should be sealed with a waterproof, non-extruding compressible compound specifically recommended for heavy duty concrete pavement and wet environments.

Where floor slabs are tied to perimeter walls or turn-down slabs to meet structural or other construction objectives, our experience indicates differential movement between the walls and slabs will likely be observed in adjacent slab expansion joints or floor slab cracks beyond the length of the structural dowels. The Structural Engineer should account for potential differential settlement through use of sufficient control joints, appropriate reinforcing or other means.

Settlement of floor slabs supported on existing fill materials cannot be accurately predicted but could be larger than normal and result in some cracking. Mitigation measures, as noted in Existing Fill within Earthwork, are critical to the performance of floor slabs. In addition to the mitigation measures, the floor slab can be stiffened by adding steel reinforcement, grade beams, and/or post-tensioned elements.

Floor Slab Construction Considerations

On most project sites, the site grading is generally accomplished early in the construction phase. However, as construction proceeds, the subgrade may be disturbed due to utility excavations, construction traffic, desiccation, rainfall, etc. As a result, the floor slab subgrade may not be suitable for placement of base rock and concrete and corrective action may be required.

We recommend the area underlying the floor slab be rough graded and then thoroughly proofrolled with a loaded tandem axle dump truck prior to final grading and placement of base

rock. Attention should be paid to high traffic areas that were rutted and disturbed earlier and to areas where backfilled trenches are located. Areas where unsuitable conditions are located should be repaired by removing and replacing the affected material with properly compacted fill. All floor slab subgrade areas should be moisture conditioned and properly compacted to the recommendations in this report immediately prior to placement of the base rock and concrete.

Finished subgrade, within and for at least 10 feet beyond the floor slab, should be protected from traffic, rutting, or other disturbance and maintained in a relatively moist condition until floor slabs are constructed. If the subgrade should become damaged or desiccated prior to construction of floor slabs, the affected material should be removed, and structural fill should be added to replace the resulting excavation. Final conditioning of the finished subgrade should be performed immediately prior to placement of the floor slab support course.

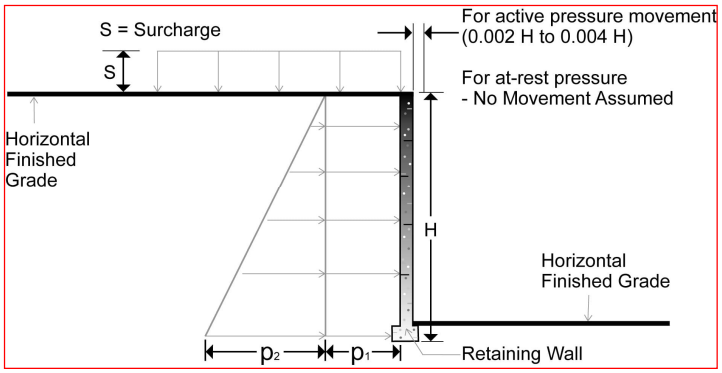
The Geotechnical Engineer should observe the condition of the floor slab subgrades immediately prior to placement of the floor slab support course, reinforcing steel, and concrete. Attention should be paid to high traffic areas that were rutted and disturbed earlier, and to areas where backfilled trenches are located.

Lateral Earth Pressures

Design Parameters

The lateral earth pressure recommendations herein are applicable to the design of rigid retaining walls subject to slight rotation, such as cantilever, or gravity type concrete walls, less than 5 feet tall. These recommendations are not applicable to the design of modular block - geogrid reinforced backfill walls. Recommendations covering these types of wall systems are beyond the scope of services for this assignment. However, we would be pleased to develop recommendations for the design of such wall systems upon request.

Structures with unbalanced backfill levels on opposite sides should be designed for earth pressures at least equal to values indicated in the following table. Earth pressures will be influenced by structural design of the walls, conditions of wall restraint, methods of construction, and/or compaction and the strength of the materials being restrained. Two wall restraint conditions are shown in the diagram below. Active earth pressure is commonly used for design of free-standing cantilever retaining walls and assumes wall movement. The “at-rest” condition assumes no wall movement and is commonly used for basement walls, loading dock walls, or other walls restrained at the top. The recommended design lateral earth pressures do not include a factor of safety and do not provide for possible hydrostatic pressure on the walls (unless stated).



Lateral Earth Pressure Design Parameters

Earth Pressure Condition	Coefficient for Backfill Type [4]	Surcharge Pressure [5,6,7] p_1 (psf)	Equivalent Fluid Pressures (psf) [4,6,7,8,9]
			Unsaturated [5]
Active (Ka) [1]	Sandy Silt / Silty Sand - 0.36	(0.36)S	(45)H
At-Rest (Ko)	Sandy Silt / Silty Sand - 0.53	(0.53)S	(65)H
Passive (Kp) [2,3]	Sandy Silt / Silty Sand – 2.8	--	--

1. For active earth pressure, wall must rotate about base, with top lateral movements 0.002 H to 0.004 H, where H is wall height.
2. For passive earth pressure, wall must move horizontally to mobilize resistance. Therefore, the total calculated passive pressure should be reduced by one-half to two-thirds for design purposes.
3. Ignore passive pressure in frost zone.
4. Uniform, horizontal backfill compacted to at least 95% of the ASTM D 698 maximum dry density, rendering a maximum unit weight of 120 pcf.
5. Uniform surcharge, where S is surcharge pressure.
6. Loading from heavy compaction equipment and dynamic loading are not included.
7. No safety factor is included in these values.
8. To achieve “Unsaturated” conditions, follow guidelines in Subsurface Drainage for Below-Grade Walls below.
9. No hydrostatic pressure acting on wall.

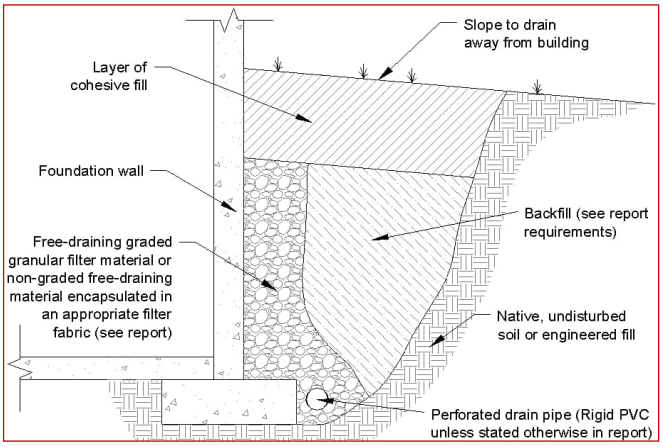
Backfill placed against structures should consist of granular soils or low plasticity cohesive soils. To calculate the resistance to sliding, a value of 0.35 should be used as the ultimate coefficient of friction between the footing and the underlying soil.

Heavy equipment should not operate within a distance closer than the exposed height of retaining walls to prevent lateral pressures more than those provided.

Footings, floor slabs or other loads bearing on backfill behind walls may have a significant influence on the lateral earth pressure. Placing footings within wall backfill and in the zone of active soil influence on the wall should be avoided unless structural analyses indicate the wall can safely withstand the increased pressure.

Subsurface Drainage for Below-Grade Walls

A perforated rigid plastic drain line installed behind the base of walls and extends below adjacent grade is recommended to prevent hydrostatic loading on the walls. The invert of a drain line around a below-grade building area or exterior retaining wall should be placed near foundation bearing level. The drain line should be sloped to provide positive gravity drainage to daylight or to a sump pit and pump. The drain line should be surrounded by clean, free-draining granular material having less than 5% passing the No. 200 sieve, such as No. 57 aggregate. The free-draining aggregate should be encapsulated in a filter fabric. The granular fill should extend to within 2 feet of final grade, where it should be capped with compacted cohesive fill to reduce infiltration of surface water into the drain system.



As an alternative to free-draining granular fill, a prefabricated drainage structure may be used. A prefabricated drainage structure is a plastic drainage core or mesh which is covered with filter fabric to prevent soil intrusion and is fastened to the wall prior to placing backfill.

Pavements

General Pavement Comments

Pavement designs are provided for the traffic conditions and pavement life conditions as noted in Project Description and in the following sections of this report. A critical

aspect of pavement performance is site preparation. Pavement designs noted in this section must be applied to the site which has been prepared as recommended in the Earthwork section.

The recommended minimum pavement sections presented below are based on the assumed traffic information along with our experience with similar developments and similar soil subgrade conditions. After traffic information has been developed, we recommend that you allow us to review the traffic data and revise our recommendations appropriately.

Please note that the recommended pavement sections are not applicable to construction traffic conditions. Any use of the pavement sections recommended (either partial or full pavement) for construction expediency (for working/storage platform) is solely at the discretion of the contractor and could require extensive remedial measures prior to final pavement construction.

Pavement Subgrade Preparation

On most project sites, the site grading is accomplished relatively early in the construction phase. Fills are placed and compacted in a uniform manner. However, as construction proceeds, excavations are made into these areas, rainfall and surface water saturates some areas, heavy traffic from concrete trucks and other delivery vehicles disturbs the subgrade and many surface irregularities are filled in with loose soils to improve trafficability temporarily. As a result, the pavement subgrades, initially prepared early in the project, should be carefully evaluated as the time for pavement construction approaches.

We recommend the moisture content and density of the top 12 inches of the subgrade be evaluated and the pavement subgrades be proofrolled within two days prior to commencement of actual paving operations. Areas not in compliance with the required ranges of moisture or density should be moisture conditioned and recompacted. Particular attention should be paid to high traffic areas that were rutted and disturbed earlier and to areas where backfilled trenches are located. Areas where unsuitable conditions are located should be repaired by removing and replacing the materials with properly compacted fills. If a significant precipitation event occurs after the evaluation or if the surface becomes disturbed, the subgrade should be reviewed by qualified personnel immediately prior to paving. The subgrade should be in its finished form at the time of the final review.

After proofrolling and repairing subgrade deficiencies, the entire subgrade should be scarified and developed as recommended in Earthwork to provide a uniform subgrade for pavement construction. Areas that appear severely desiccated following site stripping may require further undercutting and moisture conditioning.

Pavement Design Parameters

Traffic patterns and anticipated loading conditions were not available at the time that this report was prepared. However, we anticipate that traffic loads will be produced primarily by automobile traffic (350 cars per day) and occasional delivery trucks (2 trucks per day) and trash removal trucks (1 truck per week). The thickness of pavements subjected to heavy truck traffic should be determined using expected traffic volumes, vehicle types, and vehicle loads and should be in accordance with local, city or county ordinances. A pavement design life of 20 years was assumed.

An assumed California Bearing Ratio (CBR) of 4 was used for the subgrade for the asphaltic concrete (AC) pavement designs. A modulus of subgrade reaction of 100 pci was used for the Portland cement concrete (PCC) pavement designs. The values were empirically derived based upon our experience with the silty sand/sandy silt subgrade soils and our expectation of the quality of the subgrade as prescribed by the Site Preparation conditions as outlined in Earthwork. A modulus of rupture of 550 psi was used in design for the concrete (based on correlations with a minimum 28-day compressive strength of 4,000 psi).

Pavement thickness was determined using AASHTO methodology based upon the traffic loading and design parameters presented above. Terracon can provide thickness recommendations for pavements subjected to loads other than personal vehicle and occasional delivery and trash removal truck traffic if this information is provided.

The listed pavement component thicknesses should be used as a guide for pavement systems at the site for the traffic classifications and loadings stated herein. If pavement frequencies or loads will be different than that specified, Terracon should be contacted and allowed to review these pavement sections. Please note that the recommended pavement sections are not applicable to construction traffic conditions. Construction traffic on the pavements was not considered in developing the recommended minimum pavement thicknesses. If the pavements are subject to traffic by construction equipment/vehicles, the pavement thicknesses should be revised to consider the effects of the additional traffic loading. Any use of the pavement sections recommended (either partial or full pavement) for construction expediency (for working/storage platform) is solely at the discretion of the contractor and could require extensive remedial measures prior to final pavement construction.

Pavement Section Thicknesses

Asphaltic Concrete (AC) Design

The following table provides our opinion of minimum thickness for AC sections based on the traffic loading presented above. Please note that construction traffic on the

pavements was not considered in developing the recommended minimum pavement thicknesses. If the pavements are subject to traffic by construction equipment/vehicles, the pavement thicknesses should be revised to consider the effects of the additional traffic loading.

Asphaltic Concrete (AC) Design

Layer	Thickness (inches)		Reference
	Light Duty [1]	Heavy Duty [2]	
AC Surface	2	1 ¼	SP9.5 – GDOT Sections 400, 424, 824, and 828
AC Binder	--	1 ¾	SP19 – GDOT Sections 400, 424, 824, and 828
Aggregate Base	6	8	GAB, GDOT Sections 815 and 310
Subgrade	Upper 12 inches of existing soil or engineered fill	Upper 12 inches of existing soil or engineered fill	98% of Standard Proctor MDD, -2 to +3% OMC

1. Automobiles only.
2. Combined automobiles and trucks.

The graded aggregate base should be compacted to a minimum of 98 percent of the material's modified Proctor (ASTM D-1557, Method C) maximum dry density. Where base course thickness exceeds 6 inches, the material should be placed and compacted in two or more lifts of equal thickness.

Asphalt concrete aggregates and base course materials should conform to the Georgia Department of Transportation (GDOT) "Standard Specifications for Construction of Transportation System".

Portland Cement Concrete (PCC) Design

The following table provides our estimated minimum thickness of PCC pavements based on the traffic loading presented above. Please note that construction traffic on the pavements was not considered in developing the recommended minimum pavement thicknesses. If the pavements are subject to traffic by construction equipment/vehicles, the pavement thicknesses should be revised to consider the effects of the additional traffic loading.

Portland Cement Concrete (PCC) Design

Layer	Thickness (inches)		Reference
	Light Duty [1]	Heavy Duty [2]	
PCC	5	6 ½	ACI
Aggregate Base	4	4	GAB, GDOT Sections 815 and 310
Subgrade	Upper 12 inches of existing soil or engineered fill	Upper 12 inches of existing soil or engineered fill	98% of Standard Proctor MDD, -2 to +3% OMC

1. Automobiles only.
2. Combined automobiles and trucks.

We recommend a Portland cement concrete (PCC) pavement be utilized in entrance and exit sections, dumpster pads, loading dock areas, or other areas where extensive wheel maneuvering are expected. The dumpster pad should be large enough to support the wheels of the truck which will bear the load of the dumpster.

Areas for parking of heavy vehicles, concentrated turn areas, and start/stop maneuvers could require thicker pavement sections. Edge restraints (i.e. concrete curbs or aggregate shoulders) should be planned along curves and areas of maneuvering vehicles.

Although not required for structural support, the base course layer is recommended to help reduce potential for slab curl, shrinkage cracking, and subgrade pumping through joints. Proper joint spacing will also be required to prevent excessive slab curling and shrinkage cracking. Joints should be sealed to prevent entry of foreign material and doweled where necessary for load transfer. PCC pavement details for joint spacing, joint reinforcement, and joint sealing should be prepared in accordance with ACI 330 and ACI 325.

Where practical, we recommend early-entry cutting of crack-control joints in PCC pavements. Cutting of the concrete in its “green” state typically reduces the potential for micro-cracking of the pavements prior to the crack control joints being formed, compared to cutting the joints after the concrete has fully set. Micro-cracking of pavements may lead to crack formation in locations other than the sawed joints, and/or reduction of fatigue life of the pavement.

Openings in pavements, such as decorative landscaped areas, are sources for water infiltration into surrounding pavement systems. Water can collect in the islands and migrate into the surrounding subgrade soils thereby degrading support of the pavement. Islands with raised concrete curbs, irrigated foliage, and low permeability near-surface

soils are particular areas of concern. The civil design for the pavements with these conditions should include features to restrict or collect and discharge excess water from the islands. Examples of features are edge drains connected to the stormwater collection system, longitudinal subdrains, or other suitable outlets and impermeable barriers preventing lateral migration of water such as a cutoff wall installed to a depth below the pavement structure.

Pavement Drainage

Pavements should be sloped to provide rapid drainage of surface water. Water allowed to pond on or adjacent to the pavements could saturate the subgrade and contribute to premature pavement deterioration. In addition, the pavement subgrade should be graded to provide positive drainage within the granular base section.

We recommend drainage be included at the bottom of the GAB layer at the storm structures to aid in removing water that may enter this layer. Drainage could consist of small diameter weep holes excavated around the perimeter of the storm structures. The weep holes should be excavated at the elevation of the GAB and soil interface. The excavation should be covered with No. 57 stone which is encompassed in Mirafi 140 NL or approve equivalent which will aid in reducing fines from entering the storm system.

Pavement Maintenance

The pavement sections represent minimum recommended thicknesses and, as such, periodic upkeep should be anticipated. Preventive maintenance should be planned and provided for through an on-going pavement management program. Maintenance activities are intended to slow the rate of pavement deterioration and to preserve the pavement investment. Pavement care consists of both localized (e.g., crack and joint sealing and patching) and global maintenance (e.g., surface sealing). Additional engineering consultation is recommended to determine the type and extent of a cost-effective program. Even with periodic maintenance, some movements and related cracking may still occur, and repairs may be required.

Pavement performance is affected by its surroundings. In addition to providing preventive maintenance, the civil engineer should consider the following recommendations in the design and layout of pavements:

- Final grade adjacent to paved areas should slope down from the edges at a minimum 2%.
- Subgrade and pavement surfaces should have a minimum 2% slope to promote proper surface drainage.
- Install pavement drainage systems surrounding areas anticipated for frequent wetting.

- Install joint sealant and seal cracks immediately.
- Seal all landscaped areas in or adjacent to pavements to reduce moisture migration to subgrade soils.
- Place compacted, low permeability backfill against the exterior side of curb and gutter.
- Place curb, gutter and/or sidewalk directly on clay subgrade soils rather than on unbound granular base course materials.

General Comments

Our analysis and opinions are based upon our understanding of the project, the geotechnical conditions in the area, and the data obtained from our site exploration. Variations will occur between exploration point locations or due to the modifying effects of construction or weather. The nature and extent of such variations may not become evident until during or after construction. Terracon should be retained as the Geotechnical Engineer, where noted in this report, to provide observation and testing services during pertinent construction phases. If variations appear, we can provide further evaluation and supplemental recommendations. If variations are noted in the absence of our observation and testing services on-site, we should be immediately notified so that we can provide evaluation and supplemental recommendations.

Our Scope of Services does not include either specifically or by implication any environmental or biological (e.g., mold, fungi, bacteria) assessment of the site or identification or prevention of pollutants, hazardous materials or conditions. If the owner is concerned about the potential for such contamination or pollution, other studies should be undertaken.

Our services and any correspondence are intended for the sole benefit and exclusive use of our client for specific application to the project discussed and are accomplished in accordance with generally accepted geotechnical engineering practices with no third-party beneficiaries intended. Any third-party access to services or correspondence is solely for information purposes to support the services provided by Terracon to our client. Reliance upon the services and any work product is limited to our client and is not intended for third parties. Any use or reliance of the provided information by third parties is done solely at their own risk. No warranties, either express or implied, are intended or made.

Site characteristics as provided are for design purposes and not to estimate excavation cost. Any use of our report in that regard is done at the sole risk of the excavating cost estimator as there may be variations on the site that are not apparent in the data that could significantly affect excavation cost. Any parties charged with estimating excavation costs should seek their own site characterization for specific purposes to obtain the specific level of detail necessary for costing. Site safety and cost estimating including

excavation support and dewatering requirements/design are the responsibility of others. Construction and site development have the potential to affect adjacent properties. Such impacts can include damages due to vibration, modification of groundwater/surface water flow during construction, foundation movement due to undermining or subsidence from excavation, as well as noise or air quality concerns. Evaluation of these items on nearby properties are commonly associated with contractor means and methods and are not addressed in this report. The owner and contractor should consider a preconstruction/precondition survey of surrounding development. If changes in the nature, design, or location of the project are planned, our conclusions and recommendations shall not be considered valid unless we review the changes and either verify or modify our conclusions in writing.

Attachments

Exploration and Testing Procedures

Field Exploration

Summary of Field Exploration

Number of Borings	Approximate Boring Depth (feet)	Location
3	20	Building area
1	20	Banking Lane
4	10	Parking/driveway areas

Boring Layout and Elevations: Terracon personnel provided the boring layout using handheld GPS equipment (estimated horizontal accuracy of about ±10 feet) and referencing existing site features. If elevations and a more precise boring layout are desired, we recommend borings be surveyed.

Subsurface Exploration Procedures: We advanced the borings with an ATV-mounted rotary drill rig using continuous flight augers (hollow stem). Four samples were obtained in the upper 10 feet of each boring and at intervals of 5 feet thereafter. In the split-barrel sampling procedure, a standard 2-inch outer diameter split-barrel sampling spoon was driven into the ground by a 140-pound safety hammer falling a distance of about 30 inches. The number of blows required to advance the sampling spoon the last 12 inches of a normal 18-inch penetration is recorded as the Standard Penetration Test (SPT) resistance value. The SPT resistance values, also referred to as N-values, are indicated on the boring logs at the test depths. We observed and recorded groundwater levels during drilling and sampling. For safety purposes, all borings were backfilled with auger cuttings after their completion. Pavements were patched with cold-mix asphalt, as appropriate.

The sampling depths, penetration distances, and other sampling information was recorded on the field boring logs. The samples were placed in appropriate containers and taken to our soil laboratory for testing and classification by a Geotechnical Engineer. Our exploration team prepared field boring logs as part of the drilling operations. These field logs included visual classifications of the materials observed during drilling and our interpretation of the subsurface conditions between samples. Final boring logs were prepared from the field logs. The final boring logs represent the Geotechnical Engineer's interpretation of the field logs and include modifications based on observations and tests of the samples in our laboratory.

Laboratory Testing

The project engineer reviewed the field data and assigned laboratory tests. The laboratory testing program included the following types of tests:

- Moisture content
- Atterberg limits
- Grain size distribution

The laboratory testing program often included examination of soil samples by an engineer. Based on the results of our field and laboratory programs, we described and classified the soil samples in accordance with the Unified Soil Classification System.

Site Location and Exploration Plans

Contents:

Site Location Plan
Exploration Plan

Note: All attachments are one page unless noted above.

Site Location



DIAGRAM IS FOR GENERAL LOCATION ONLY, AND IS NOT INTENDED FOR CONSTRUCTION PURPOSES MAP PROVIDED BY MICROSOFT BING MAPS

Exploration Plan

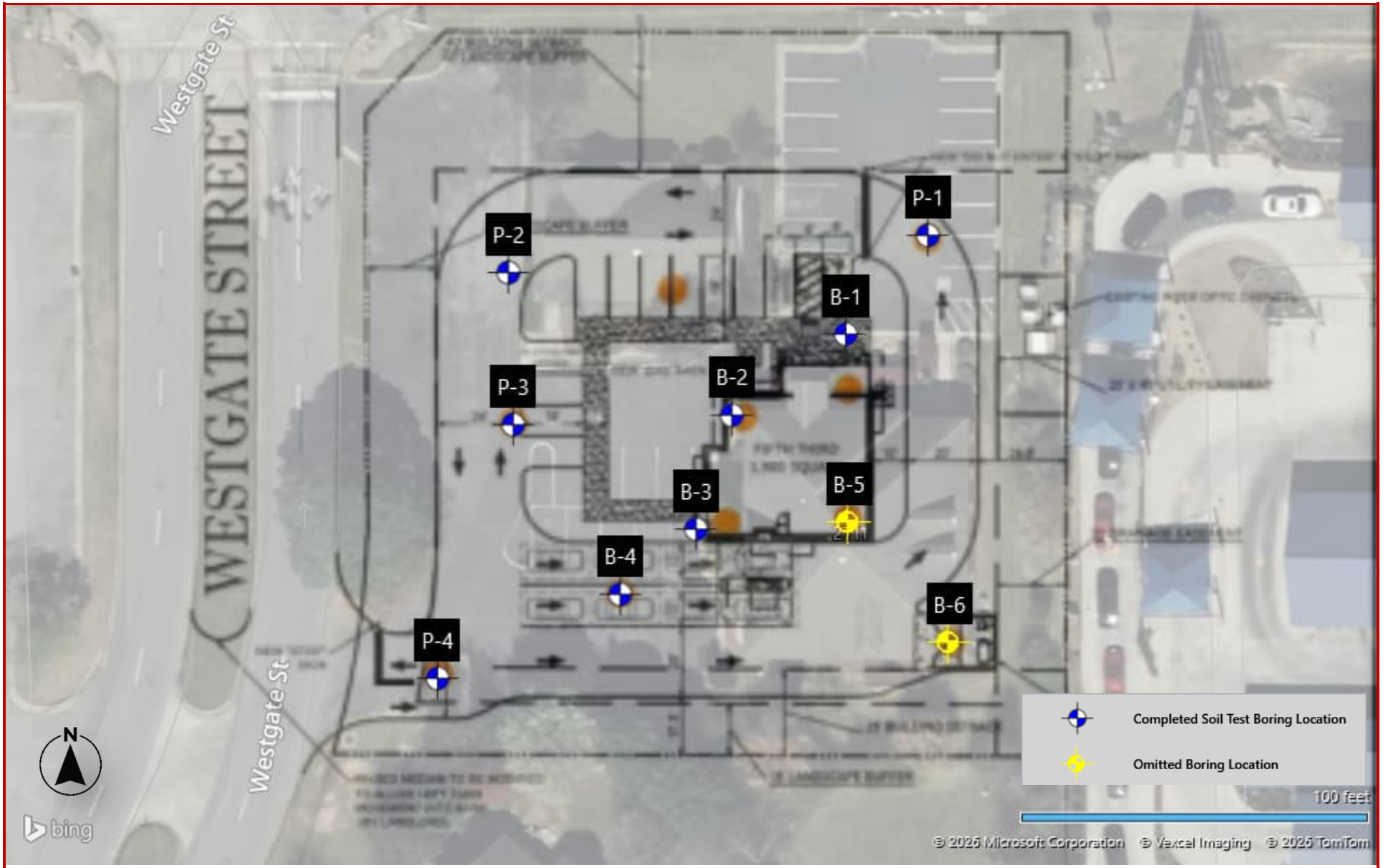


DIAGRAM IS FOR GENERAL LOCATION ONLY, AND IS NOT INTENDED FOR CONSTRUCTION PURPOSES

MAP PROVIDED BY MICROSOFT BING MAPS

Exploration and Laboratory Results

Contents:

Boring Logs (B-1 through B-4 and P-1 through P-4)
Moisture Content (3 pages)
Atterberg Limits (3 pages)
Grain Size Distribution (3 pages)
GeoModel

Note: All attachments are one page unless noted above.

Fifth Third Bank - Watson
2711 Watson Boulevard, Warner Robins, GA,USA
Terracon Project No. 49265099

Latitude: 32.6179° Longitude: -83.6713°

BORING LOG NO. B-1

Terracon
2105 Newpoint Pl, Ste 600
Lawrenceville, GA 30043-5557

Model Layer	Graphic Log	Lithology Depth (Ft.)	Material Description	Depth (Ft.)	Elevation (Ft.)	Sample Type	Recovery (In.)	Field Test Results	Water Content (%)	Percent Gravel	Percent Sand	Percent Fines	Atterberg Limits LL	PL	PI
1		0.2	**ASPHALT,** 2 Inches												
		1.2	**AGGREGATE BASE COURSE,** 12 Inches												
2			**FILL- SANDY LEAN CLAY,** fine grained sand, brown, moist, stiff				18	3-4-6 N = 10							
		3.0													
3			**SANDY LEAN CLAY (CL),** fine grained sand, brown, moist, stiff, coastal plain				18	4-4-5 N = 9	21	0	47	53	37	19	18
				5											
		6.0	very stiff				18	5-7-9 N = 16							
		8.0	**CLAYEY SAND (SC),** fine grained sand, brown, moist, dense, coastal plain				18	12-15-20 N = 35							
				10											
							18	10-14-20 N = 34							
				15											
							18	17-18-20 N = 38							
			Boring Terminated at 20 Ft												

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Notes

Water Level Observations
Groundwater not encountered while drilling

Advancement Method
0-20 Ft. Hollow Stem Auger

Abandonment Method
Boring backfilled with soil cuttings upon completion.

Equipment
604/CME-45B

Hammer Type
Rope and Cathead

Driller
Harddison Drilling - Cody

Logged By
Ethan Fraser

Start Date
04/17/2026

Completion Date
04/17/2026

Fifth Third Bank - Watson
2711 Watson Boulevard, Warner Robins, GA,USA
Terracon Project No. 49265099

Latitude: 32.6178° Longitude: -83.6714°

BORING LOG NO. B-2

Terracon
2105 Newpoint Pl, Ste 600
Lawrenceville, GA 30043-5557

Model Layer	Graphic Log	Lithology Depth (Ft.)	Material Description	Depth (Ft.)	Elevation (Ft.)	Sample Type	Recovery (In.)	Field Test Results
1		0.2	**ASPHALT,** 2 Inches					
		0.7	**AGGREGATE BASE COURSE,** 6 Inches					
3			**CLAYEY SAND (SC),** fine grained sand, orange-brown, moist, medium dense, coastal plain					
							18	5-9-13 N = 22
							18	5-8-8 N = 16
				5			18	5-7-11 N = 18
							18	5-9-12 N = 21
				10			18	7-11-13 N = 24
				15			18	11-14-15 N = 29
			Boring Terminated at 20 Ft					

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Notes

Water Level Observations
Groundwater not encountered while drilling

Advancement Method
0-20 Ft. Hollow Stem Auger

Abandonment Method
Boring backfilled with soil cuttings upon completion.

Equipment
604/CME-45B

Hammer Type
Rope and Cathead

Driller
Harddison Drilling - Cody

Logged By
Ethan Fraser

Start Date
04/17/2026

Completion Date
04/17/2026

Fifth Third Bank - Watson
2711 Watson Boulevard, Warner Robins, GA,USA
Terracon Project No. 49265099

Latitude: 32.6177° Longitude: -83.6715°

BORING LOG NO. B-3

Terracon
2105 Newpoint Pl, Ste 600
Lawrenceville, GA 30043-5557

Model Layer	Graphic Log	Lithology Depth (Ft.)	Material Description	Depth (Ft.)	Elevation (Ft.)	Sample Type	Recovery (In.)	Field Test Results	Water Content (%)	Percent Gravel	Percent Sand	Percent Fines	Atterberg Limits LL	Atterberg Limits PL	Atterberg Limits PI
1		0.3	**ASPHALT,** 3 Inches												
		1.3	**AGGREGATE BASE COURSE,** 12 Inches												
3			**SANDY LEAN CLAY (CL),** fine grained sand, orange-brown, moist, very stiff, coastal plain				18	4-8-11 N = 19	21	0.9	39.1	60	41	18	23
							18	5-8-9 N = 17							
				5			18	6-11-14 N = 25							
		8.0	**CLAYEY SAND (SC),** fine grained sand, red-brown, moist, dense, coastal plain				18	8-15-16 N = 31							
				10											
							18	12-16-19 N = 35							
				15											
							18	10-15-17 N = 32							
			Boring Terminated at 20 Ft												

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Notes

Water Level Observations
Groundwater not encountered while drilling

Advancement Method
0-20 Ft. Hollow Stem Auger

Abandonment Method
Boring backfilled with soil cuttings upon completion.

Equipment
604/CME-45B

Hammer Type
Rope and Cathead

Driller
Harddison Drilling - Cody

Logged By
Ethan Fraser

Start Date
04/17/2026

Completion Date
04/17/2026

Fifth Third Bank - Watson
2711 Watson Boulevard, Warner Robins, GA,USA
Terracon Project No. 49265099

Latitude: 32.6177° Longitude: -83.6716°

BORING LOG NO. B-4

Terracon
2105 Newpoint Pl, Ste 600
Lawrenceville, GA 30043-5557

Model Layer	Graphic Log	Lithology Depth (Ft.)	Material Description	Depth (Ft.)	Elevation (Ft.)	Sample Type	Recovery (In.)	Field Test Results
1		0.2	**ASPHALT,** 2 Inches					
		0.7	**AGGREGATE BASE COURSE,** 6 Inches					
3			**SANDY LEAN CLAY (CL),** fine grained sand, orange-brown, moist, very stiff, coastal plain				18	4-7-11 N = 18
		3.5	stiff				18	4-7-8 N = 15
		6.0	very stiff	5			18	5-6-10 N = 16
		8.0	**CLAYEY SAND (SC),** fine grained sand, orange-brown, red, moist, medium dense, coastal plain				18	4-8-13 N = 21
				10			18	8-11-15 N = 26
				15			18	10-13-15 N = 28
			Boring Terminated at 20 Ft					

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Notes

Water Level Observations
Groundwater not encountered while drilling

Advancement Method
0-20 Ft. Hollow Stem Auger

Abandonment Method
Boring backfilled with soil cuttings upon completion.

Equipment
604/CME-45B

Hammer Type
Rope and Cathead

Driller
Harddison Drilling - Cody

Logged By
Ethan Fraser

Start Date
04/17/2026

Completion Date
04/17/2026

Fifth Third Bank - Watson
2711 Watson Boulevard, Warner Robins, GA,USA
Terracon Project No. 49265099

Latitude: 32.6180° Longitude: -83.6713°

BORING LOG NO. P-1

2105 Newpoint Pl, Ste 600
Lawrenceville, GA 30043-5557

Model Layer	Graphic Log	Lithology Depth (Ft.)	Material Description	Depth (Ft.)	Elevation (Ft.)	Sample Type	Recovery (In.)	Field Test Results
1		0.2	**ASPHALT,** 2 Inches					
		0.7	**AGGREGATE BASE COURSE,** 6 Inches					
2			**FILL- SANDY LEAN CLAY,** fine grained sand, brown, moist, very stiff				18	4-6-13 N = 19
		3.0						
3			**SANDY LEAN CLAY (CL),** fine grained sand, brown, moist, very stiff, coastal plain				18	5-7-11 N = 18
				5				
							18	7-10-13 N = 23
		8.0						
			CLAYEY SAND (SC), fine grained sand, brown, moist, medium dense, coastal plain				18	7-9-15 N = 24
			Boring Terminated at 10 Ft					

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Notes

Water Level Observations
Groundwater not encountered while drilling

Advancement Method
0-10 Ft. Hollow Stem Auger

Abandonment Method
Boring backfilled with soil cuttings upon completion.

Equipment
604/CME-45B

Hammer Type
Rope and Cathead

Driller
Harddison Drilling - Cody

Logged By
Ethan Fraser

Start Date
04/17/2026

Completion Date
04/17/2026

Fifth Third Bank - Watson
2711 Watson Boulevard, Warner Robins, GA,USA
Terracon Project No. 49265099

Latitude: 32.6179° Longitude: -83.6717°

BORING LOG NO. P-2

Terracon
2105 Newpoint Pl, Ste 600
Lawrenceville, GA 30043-5557

Model Layer	Graphic Log	Lithology Depth (Ft.)	Material Description	Depth (Ft.)	Elevation (Ft.)	Sample Type	Recovery (In.)	Field Test Results
1		0.2	**ASPHALT,** 2 Inches					
2			**FILL- SANDY LEAN CLAY,** fine grained sand, orange-brown, moist, firm					
							18	3-3-3 N = 6
		3.0						
3			**CLAYEY SAND (SC),** fine grained sand, orange-brown, red, moist, medium dense, coastal plain					
							18	3-6-8 N = 14
				5				
							18	8-11-13 N = 24
							18	9-12-17 N = 29
			Boring Terminated at 10 Ft					

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Notes

Water Level Observations
Groundwater not encountered while drilling

Advancement Method
0-10 Ft. Hollow Stem Auger

Abandonment Method
Boring backfilled with soil cuttings upon completion.

Equipment
604/CME-45B

Hammer Type
Rope and Cathead

Driller
Harddison Drilling - Cody

Logged By
Ethan Fraser

Start Date
04/17/2026

Completion Date
04/17/2026

Fifth Third Bank - Watson
2711 Watson Boulevard, Warner Robins, GA,USA
Terracon Project No. 49265099

Latitude: 32.6178° Longitude: -83.6717°

BORING LOG NO. P-3

Terracon
2105 Newpoint Pl, Ste 600
Lawrenceville, GA 30043-5557

Model Layer	Graphic Log	Lithology Depth (Ft.)	Material Description	Depth (Ft.)	Elevation (Ft.)	Sample Type	Recovery (In.)	Field Test Results	Water Content (%)	Percent Gravel	Percent Sand	Percent Fines	Atterberg Limits LL	PL	PI
1		0.2	**ASPHALT,** 2 Inches												
2			**FILL- SANDY LEAN CLAY,** fine grained sand, red-brown, moist, stiff				18	3-5-6 N = 11							
3		3.0	**SANDY LEAN CLAY (CL),** fine grained sand, red-brown, orange, moist, stiff, coastal plain				18	4-7-8 N = 15	21	0.3	48.4	51.3	38	18	20
				5											
		6.0	**CLAYEY SAND (SC),** fine grained sand, red-brown, orange, moist, medium dense, coastal plain				18	7-11-15 N = 26							
							18	10-12-15 N = 27							
			Boring Terminated at 10 Ft												

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Notes

Water Level Observations
Groundwater not encountered while drilling

Advancement Method
0-10 Ft. Hollow Stem Auger

Abandonment Method
Boring backfilled with soil cuttings upon completion.

Equipment
604/CME-45B

Hammer Type
Rope and Cathead

Driller
Harddison Drilling - Cody

Logged By
Ethan Fraser

Start Date
04/17/2026

Completion Date
04/17/2026

Fifth Third Bank - Watson
2711 Watson Boulevard, Warner Robins, GA,USA
Terracon Project No. 49265099

Latitude: 32.6176° Longitude: -83.6717°

BORING LOG NO. P-4

2105 Newpoint Pl, Ste 600
Lawrenceville, GA 30043-5557

Model Layer	Graphic Log	Lithology Depth (Ft.)	Material Description	Depth (Ft.)	Elevation (Ft.)	Sample Type	Recovery (In.)	Field Test Results
1		0.3	**ASPHALT,** 3 Inches					
2			**FILL- SANDY LEAN CLAY,** fine grained sand, orange-brown, moist, stiff					
							18	3-5-8 N = 13
		3.0						
3			**SANDY LEAN CLAY (CL),** fine grained sand, orange-brown, red, moist, stiff, coastal plain					
							18	4-5-10 N = 15
				5				
		6.0						
			very stiff				18	6-10-10 N = 20
							18	8-12-13 N = 25
			Boring Terminated at 10 Ft					

See Exploration and Testing Procedures for a description of field and laboratory procedures used and additional data (If any).

See Supporting Information for explanation of symbols and abbreviations.

Notes

Water Level Observations
Groundwater not encountered while drilling

Advancement Method
0-10 Ft. Hollow Stem Auger

Abandonment Method
Boring backfilled with soil cuttings upon completion.

Equipment
604/CME-45B

Hammer Type
Rope and Cathead

Driller
Harddison Drilling - Cody

Logged By
Ethan Fraser

Start Date
04/17/2026

Completion Date
04/17/2026

Fifth Third Bank - Watson
2711 Watson Boulevard | Warner Robins, GA
Terracon Project No. 49265099

Terracon
Lawrenceville, GA 30043-5557

Laboratory Determination of Water (Moisture) Content of Soil and Rock

ASTM D2216

Boring ID	B-1
Depth (ft.)	3.5 - 5
Method Used	B
Water Content (%)	20.5
Visual Description	

Fifth Third Bank - Watson
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Terracon Project No. 49265099

Terracon
Lawrenceville, GA 30043-5557

Laboratory Determination of Water (Moisture) Content of Soil and Rock

ASTM D2216

Boring ID	B-3
Depth (ft.)	1 - 2.5
Method Used	B
Water Content (%)	20.6
Visual Description	

Fifth Third Bank - Watson
2711 Watson Boulevard | Warner Robins, GA
Terracon Project No. 49265099

Terracon
Lawrenceville, GA 30043-5557

Laboratory Determination of Water (Moisture) Content of Soil and Rock

ASTM D2216

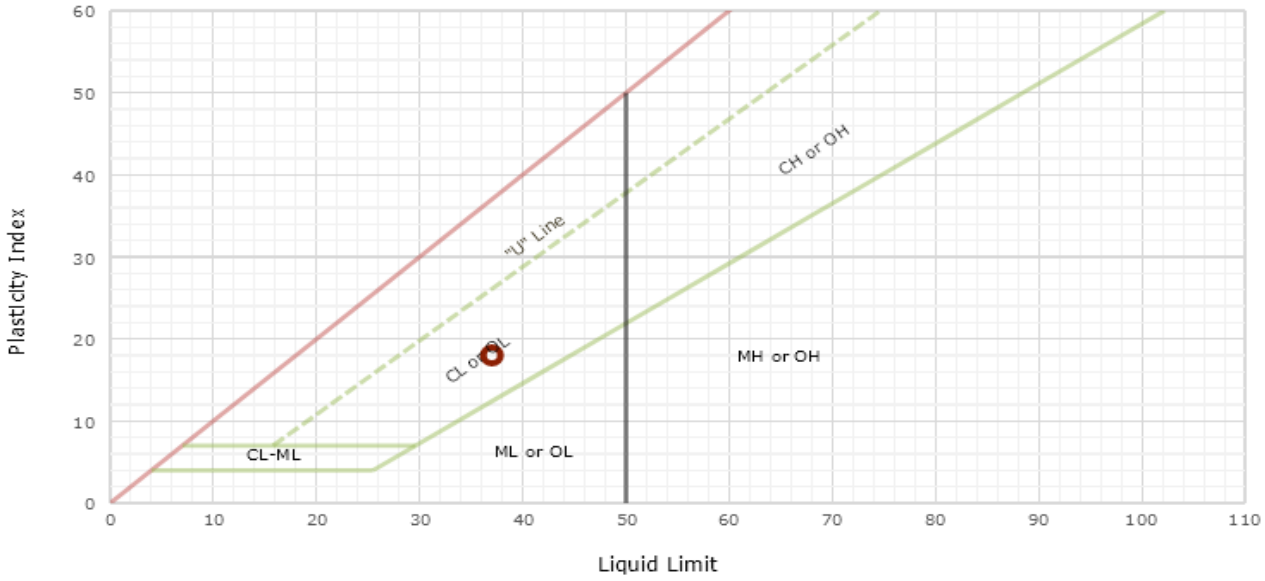
Boring ID	P-3
Depth (ft.)	3.5 - 5
Method Used	B
Water Content (%)	20.6
Visual Description	

Fifth Third Bank - Watson
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Terracon Project No. 49265099

Terracon
Lawrenceville, GA 30043-5557

Liquid Limit, Plastic Limit and Plasticity Index of Soils

ASTM D4318



Boring ID	Depth (Ft)	LL	PL	PI	Fines (%)	Description	USCS
B-1	3.5-5	37	19	18	53	Sandy Lean Clay	CL

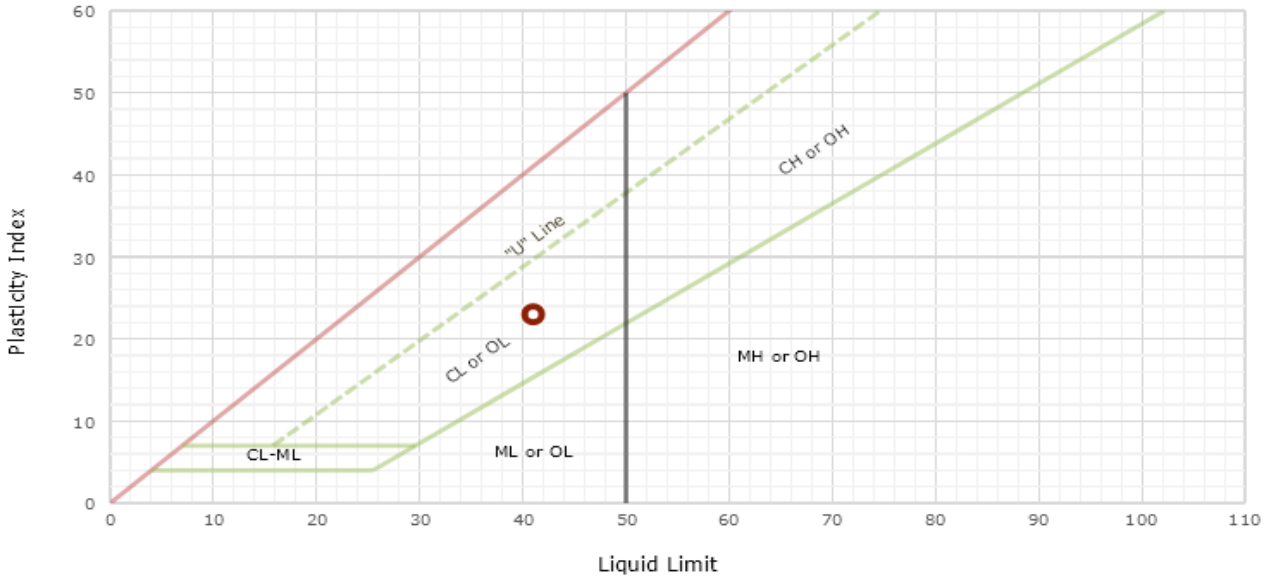
Remarks	

Fifth Third Bank - Watson
2711 Watson Boulevard | Warner Robins, GA
Terracon Project No. 49265099

Terracon
Lawrenceville, GA 30043-5557

Liquid Limit, Plastic Limit and Plasticity Index of Soils

ASTM D4318



Boring ID	Depth (Ft)	LL	PL	PI	Fines (%)	Description	USCS
B-3	1-2.5	41	18	23	60	Sandy Lean Clay	CL

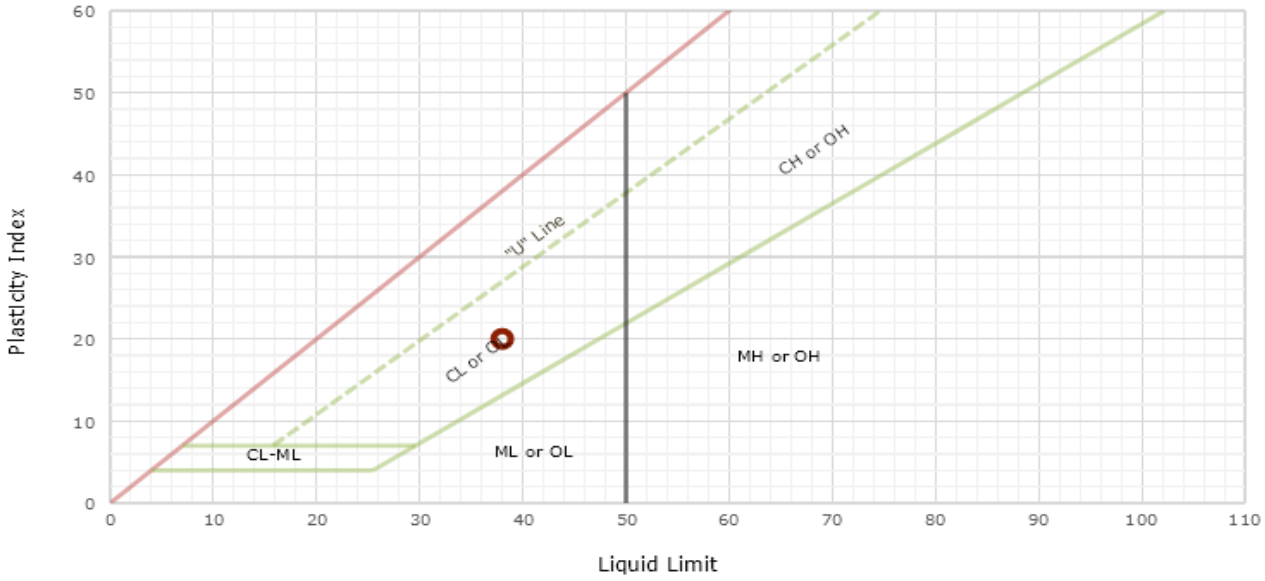
Remarks	

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Terracon Project No. 49265099

Terracon
Lawrenceville, GA 30043-5557

Liquid Limit, Plastic Limit and Plasticity Index of Soils

ASTM D4318



Boring ID	Depth (Ft)	LL	PL	PI	Fines (%)	Description	USCS
P-3	3.5-5	38	18	20	51	Sandy Lean Clay	CL

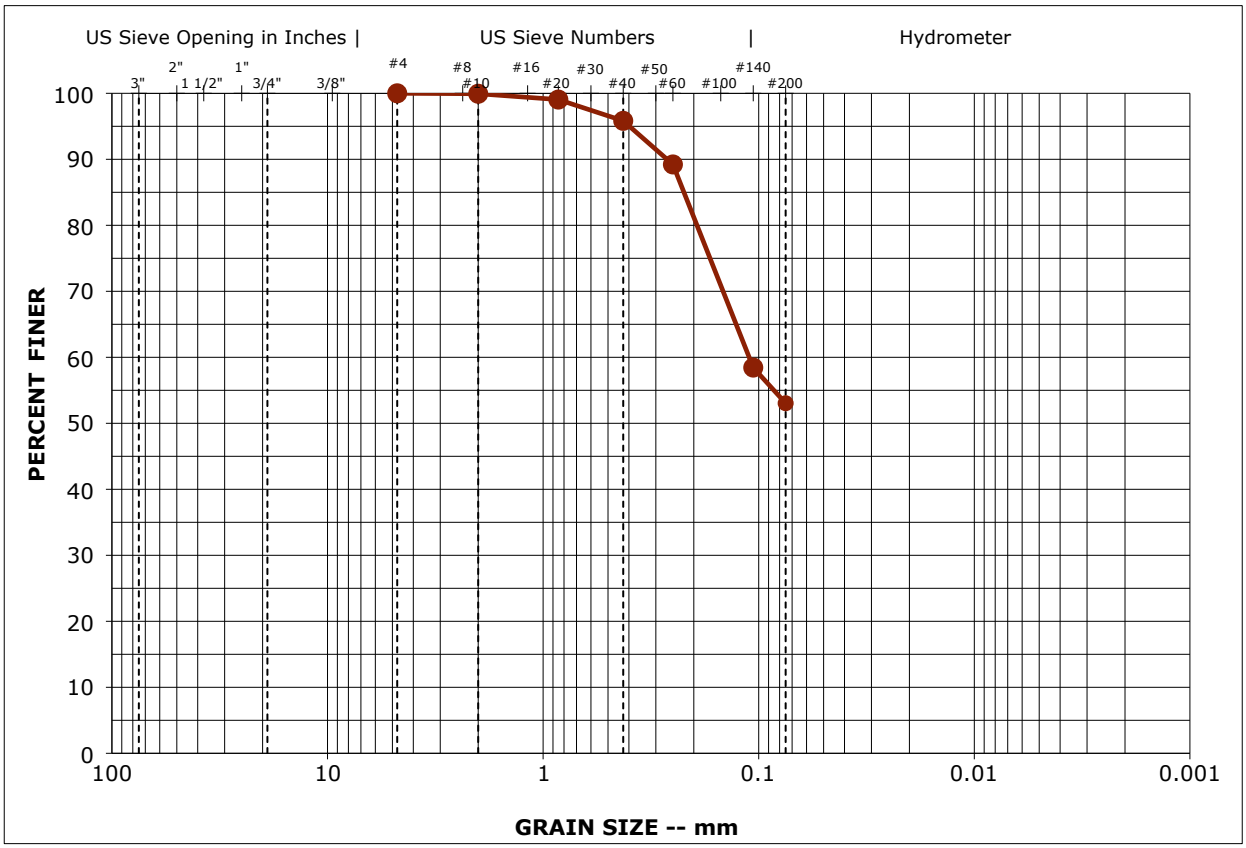
Remarks	

Fifth Third Bank - Watson
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Terracon
Lawrenceville, GA 30043-5557

Particle-Size Distribution of Soils Using Sieve Analysis

ASTM D422-Method B



Boring ID	Depth (Ft.)	Description	USCS	Specific Gravity
B-1	3.5 - 5	Sandy Lean Clay	CL	N/A

% Cobbles	% Gravel	% Sand	% Fines	% Silt	% Clay
0.0	0.0	47.0	53.0	N/A	N/A

Sieve	% Finer	Sieve	% Finer
3"	100.0	#40	95.8
2"	100.0	#60	89.2
1 1/2"	100.0	#140	58.4
1"	100.0	#200	53.0
3/4"	100.0		
3/8"	100.0		
#4	100.0		
#10	99.9		
#20	99.1		

Grain Size		Coefficients	
D_{100}	4.750	C_C	
D_{60}	0.111		
D_{30}		C_U	
D_{10}			

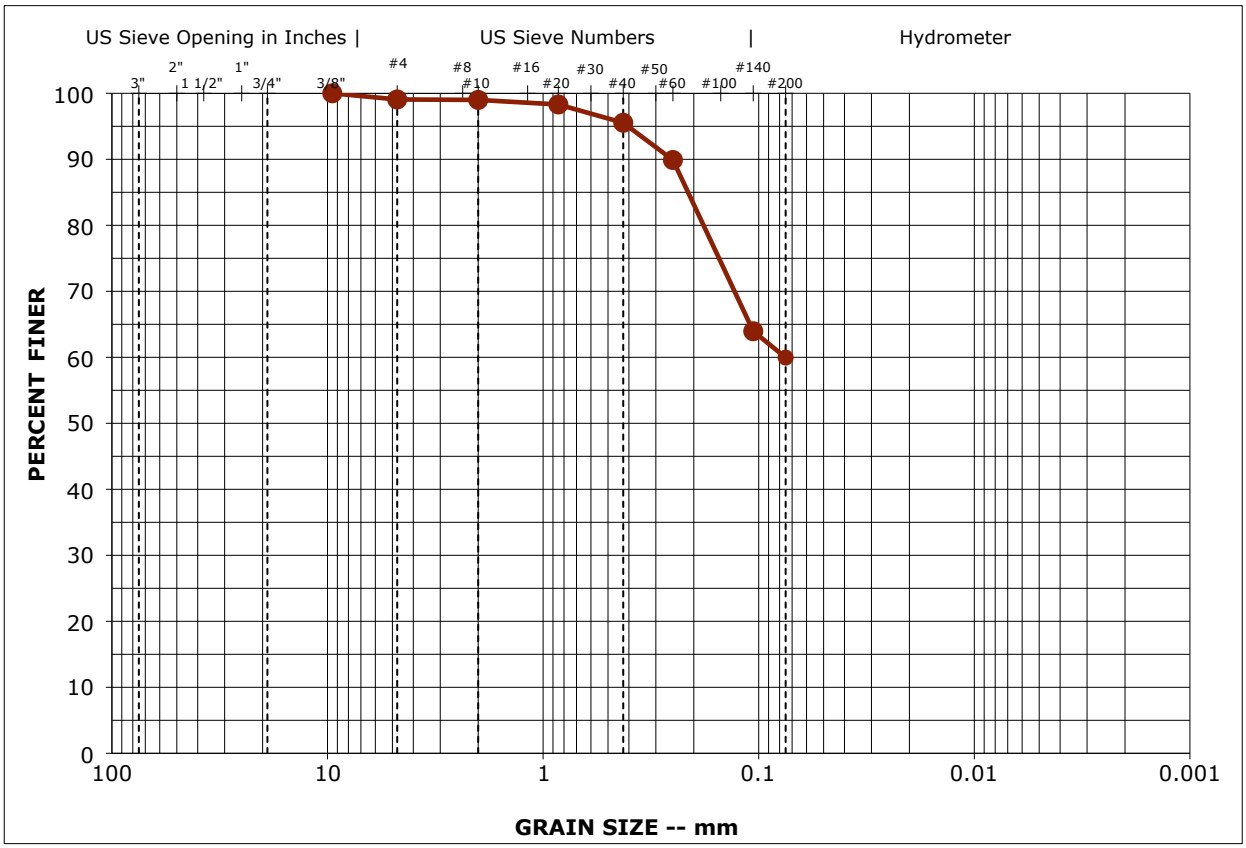
Remarks

Fifth Third Bank - Watson
2711 Watson Boulevard | Warner Robins, GA
Terracon Project No. 49265099

Terracon
Lawrenceville, GA 30043-5557

Particle-Size Distribution of Soils Using Sieve Analysis

ASTM D422-Method B



Boring ID	Depth (Ft.)	Description	USCS	Specific Gravity
B-3	1 - 2.5	Sandy Lean Clay	CL	N/A

% Cobbles	% Gravel	% Sand	% Fines	% Silt	% Clay
0.0	0.9	39.1	60.0	N/A	N/A

Sieve	% Finer	Sieve	% Finer
3"	100.0	#40	95.5
2"	100.0	#60	89.9
1 1/2"	100.0	#140	63.9
1"	100.0	#200	60.0
3/4"	100.0		
3/8"	100.0		
#4	99.1		
#10	99.0		
#20	98.3		

Grain Size		Coefficients	
D_{100}	9.500	C_C	
D_{60}	0.075		
D_{30}		C_U	
D_{10}			

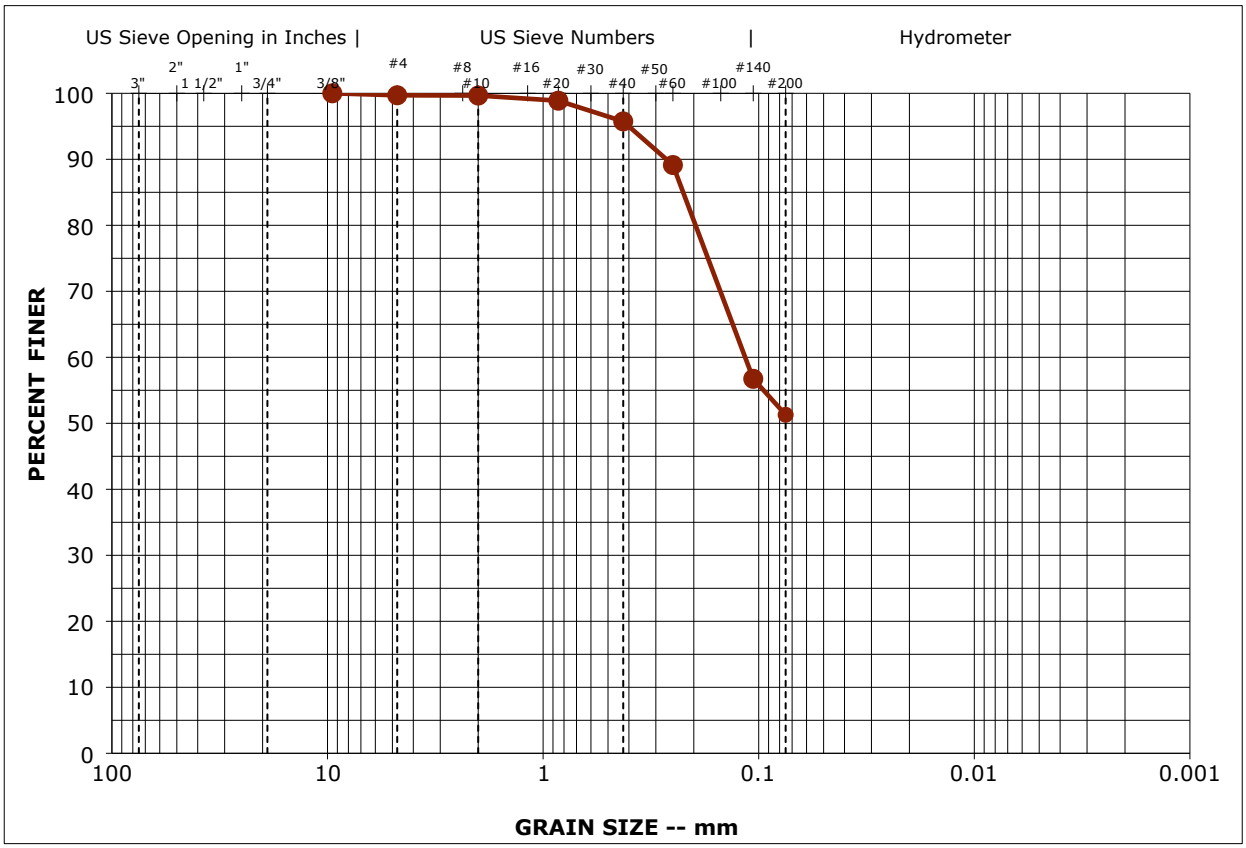
Remarks

Fifth Third Bank - Watson
2711 Watson Boulevard | Warner Robins, GA
Terracon Project No. 49265099

Terracon
Lawrenceville, GA 30043-5557

Particle-Size Distribution of Soils Using Sieve Analysis

ASTM D422-Method B



Boring ID	Depth (Ft.)	Description	USCS	Specific Gravity
P-3	3.5 - 5	Sandy Lean Clay	CL	N/A

% Cobbles	% Gravel	% Sand	% Fines	% Silt	% Clay
0.0	0.3	48.4	51.3	N/A	N/A

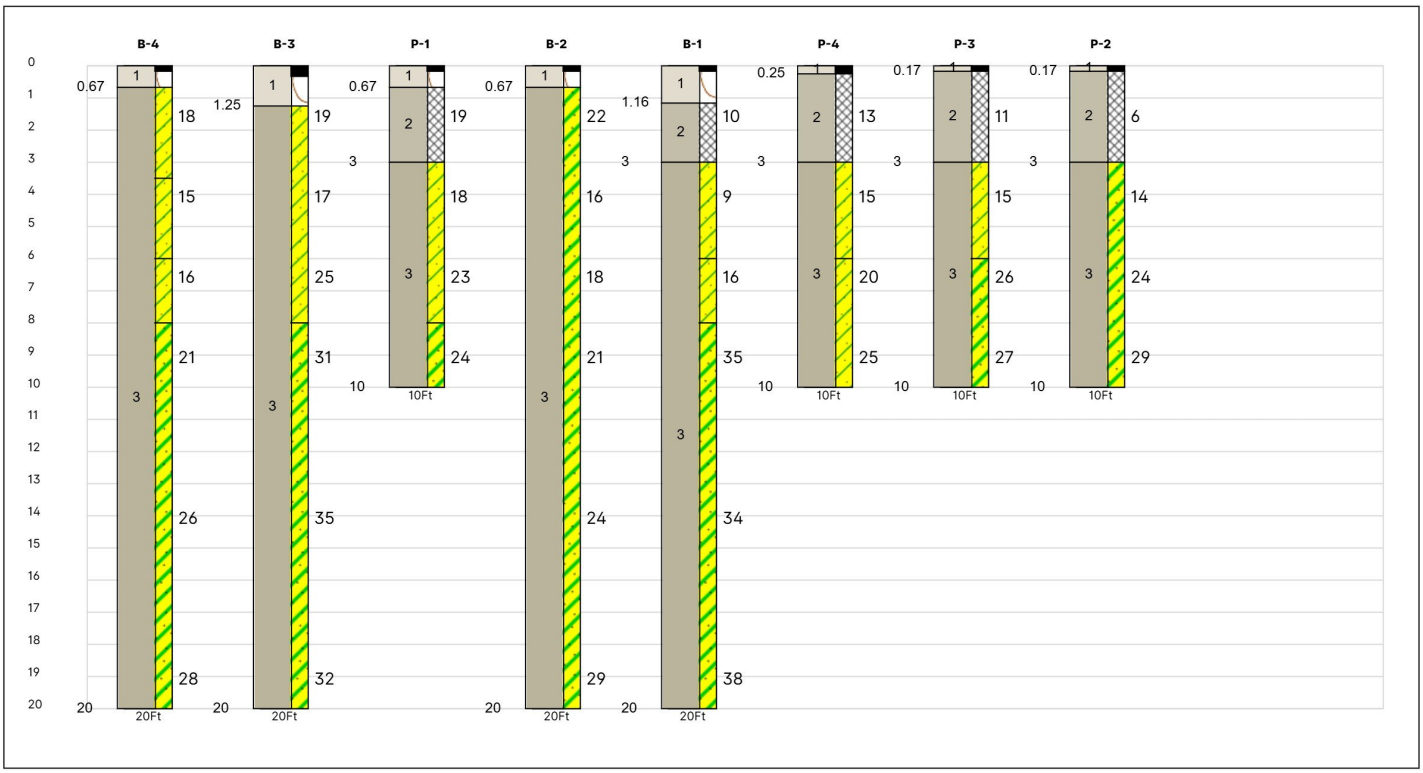
Sieve	% Finer	Sieve	% Finer
3"	100.0	#40	95.7
2"	100.0	#60	89.1
1 1/2"	100.0	#140	56.7
1"	100.0	#200	51.3
3/4"	100.0		
3/8"	100.0		
#4	99.7		
#10	99.7		
#20	98.9		

Grain Size		Coefficients	
D_{100}	9.500	C_C	
D_{60}	0.116		
D_{30}		C_U	
D_{10}			

Remarks

Terracon Project No: 49265099
Fifth Third Bank - Watson
2711 Watson Boulevard, Warner Robins, GA,USA

GeoModel



This is not a cross section. This is intended to display the Geotechnical Model only. See individual logs for more detailed conditions

#	Layer Name	General Description
1	**Surficial Materials**	Asphalt, Aggregate Base Course
2	**Fill**	Fill - Sandy Lean Clay
3	**Native Soils**	Coastal Plain - Sandy Lean Clay and Clayey Sand

Legend

Asphalt
Aggregate Base Course
Sandy Lean Clay
Clayey Sand
Fill

Groundwater levels are temporal. The levels shown are representative of the date and time of our exploration. Significant changes are possible over time.
Water levels shown are as measured during and/or after drilling. In some cases, boring advancement methods mask the presence/absence of groundwater. See individual logs for details.

First Water Observation
Second Water Observation
Third Water Observation

Notes:

Layering shown on this figure has been developed by the geotechnical engineer for purposes of modeling the subsurface conditions as required for the subsequent geotechnical engineering for this project.
Numbers adjacent to soil column indicate depth below groundsurface.

Supporting Information

Contents:

General Notes
Unified Soil Classification System

Note: All attachments are one page unless noted above.

General Notes

Sampling			Water Level	Field Tests	
Auger Cuttings	Modified California Ring Sampler	Rock Core	Water Initially Encountered	N	Standard Penetration Test Resistance (Blows/Ft.)
Dynamic Cone Penetrometer	Modified Dames & Moore Ring Sampler	Dual Sampler SPT	Water Level After a Specified Period of Time	(HP)	Hand Penetrometer
Grab Sample	GeoProbe Macro Core or Large Bore	No Recovery	Water Level After a Specified Period of Time	(T)	Torvane
Ring Sampler	Shelby Tube	Standard Penetration Test	Cave In Encountered	(DCP)	Dynamic Cone Penetrometer
Split Spoon	Texas Cone Penetrometer	Vane Shear	Water levels indicated on the soil boring logs are the levels measured in the borehole at the times indicated. Groundwater level variations will occur over time. In low permeability soils, accurate determination of groundwater levels is not possible with short term water level observations.	UC	Unconfined Compressive Strength
				(PID)	Photo-Ionization Detector
				(OVA)	Organic Vapor Analyzer

Descriptive Soil Classification

Soil classification as noted on the soil boring logs is based Unified Soil Classification System. Where sufficient laboratory data exist to classify the soils consistent with ASTM D2487 "Classification of Soils for Engineering Purposes" this procedure is used. ASTM D2488 "Description and Identification of Soils (Visual-Manual Procedure)" is also used to classify the soils, particularly where insufficient laboratory data exist to classify the soils in accordance with ASTM D2487. In addition to USCS classification, coarse grained soils are classified on the basis of their in-place relative density, and fine-grained soils are classified on the basis of their consistency. See "Strength Terms" table below for details. The ASTM standards noted above are for reference to methodology in general. In some cases, variations to methods are applied as a result of local practice or professional judgment.

Location And Elevation Notes

Exploration point locations as shown on the Exploration Plan and as noted on the soil boring logs in the form of Latitude and Longitude are approximate. See Exploration and Testing Procedures in the report for the methods used to locate the exploration points for this project. Surface elevation data annotated with +/- indicates that no actual topographical survey was conducted to confirm the surface elevation. Instead, the surface elevation was approximately determined from topographic maps of the area.

Strength Terms

Relative Density of Coarse-Grained Soils (More than 50% retained on No. 200 sieve.) Density determined by Standard Penetration Resistance		Consistency of Fine-Grained Soils (50% or more passing the No. 200 sieve.) Consistency determined by laboratory shear strength testing, field visual-manual procedures or standard penetration resistance		
Relative Density	**Standard Penetration or N-Value (Blows/Ft.)**	**Consistency**	**Unconfined Compressive Strength Qu (tsf)**	**Standard Penetration or N-Value (Blows/Ft.)**
Very Loose	0 - 3	Very Soft	less than 0.25	0 - 1
Loose	4 - 9	Soft	0.25 to 0.50	2 - 4
Medium Dense	10 - 29	Firm	0.50 to 1.00	5 - 8
Dense	30 - 50	Stiff	1.00 to 2.00	9 - 15
Very Dense	> 50	Very Stiff	2.00 to 4.00	16 - 30
		Hard	> 4.00	> 30

Relevance of Exploration and Laboratory Test Results

Exploration/field results and/or laboratory test data contained within this document are intended for application to the project as described in this document. Use of such exploration/field results and/or laboratory test data should not be used independently of this document.

Unified Soil Classification System

Criteria for Assigning Group Symbols and Group Names Using Laboratory Tests [A]				Soil Classification	
				Group Symbol	Group Name [B]
Coarse-Grained Soils: More than 50% retained on No. 4 sieve	Gravels: More than 50% of coarse fraction retained on No. 4 sieve	Clean Gravels: Less than 5% fines [C]	Cu≥4 and 1≤Cc≤3 [E]	GW	Well-graded gravel [F]
			Cu<4 and/or [Cc<1 or Cc>3.0] [E]	GP	Poorly graded gravel [F]
		Gravels with Fines: More than 12% fines [C]	Fines classify as ML or MH	GM	Silty gravel [F, G, H]
			Fines classify as CL or CH	GC	Clayey gravel [F, G, H]
	Sands: 50% or more of coarse fraction passes No. 4 sieve	Clean Sands: Less than 5% fines [D]	Cu≥6 and 1≤Cc≤3 [E]	SW	Well-graded sand [I]
			Cu<6 and/or [Cc<1 or Cc>3.0] [E]	SP	Poorly graded sand [I]
		Sands with Fines: More than 12% fines [D]	Fines classify as ML or MH	SM	Silty sand [G, H, I]
			Fines classify as CL or CH	SC	Clayey sand [G, H, I]
Fine-Grained Soils: 50% or more passes the No. 200 sieve	Silts and Clays: Liquid limit less than 50	Inorganic:	PI > 7 and plots above "A" line [J]	CL	Lean clay [K, L, M]
			PI < 4 or plots below "A" line [J]	ML	Silt [K, L, M]
		Organic:	$\frac{LL\ oven\ dried}{LL\ not\ dried}$ < 0.75	OL	Organic clay [K, L, M, N]
					Organic silt [K, L, M, O]
	Silts and Clays: Liquid limit 50 or more	Inorganic:	PI plots on or above "A" line	CH	Fat clay [K, L, M]
			PI plots below "A" line	MH	Elastic silt [K, L, M]
		Organic:	$\frac{LL\ oven\ dried}{LL\ not\ dried}$ < 0.75	OH	Organic clay [K, L, M, P]
					Organic silt [K, L, M, Q]
Highly organic soils:	Primarily organic matter, dark in color, and organic odor			PT	Peat

[A] Based on the material passing the 3-inch (75-mm) sieve.
[B] If field sample contained cobbles or boulders, or both, add "with cobbles or boulders, or both" to group name.
[C] Gravels with 5 to 12% fines require dual symbols: GW-GM well-graded gravel with silt, GW-GC well-graded gravel with clay, GP-GM poorly graded gravel with silt, GP-GC poorly graded gravel with clay.
[D] Sands with 5 to 12% fines require dual symbols: SW-SM well-graded sand with silt, SW-SC well-graded sand with clay, SP-SM poorly graded sand with silt, SP-SC poorly graded sand with clay.
[E] $Cu = D_{60}/D_{10}$ $Cc = \frac{(D_{30})^2}{D_{10} \times D_{60}}$
[F] If soil contains ≥ 15% sand, add "with sand" to group name.
[G] If fines classify as CL-ML, use dual symbol GC-GM, or SC-SM.
[H] If fines are organic, add "with organic fines" to group name.
[I] If soil contains ≥ 15% gravel, add "with gravel" to group name.
[J] If Atterberg limits plot in shaded area, soil is a CL-ML, silty clay.
[K] If soil contains 15 to 29% plus No. 200, add "with sand" or "with gravel," whichever is predominant.
[L] If soil contains ≥ 30% plus No. 200 predominantly sand, add "sandy" to group name.
[M] If soil contains ≥ 30% plus No. 200, predominantly gravel, add "gravelly" to group name.
[N] PI ≥ 4 and plots on or above "A" line.
[O] PI < 4 or plots below "A" line.
[P] PI plots on or above "A" line.
[Q] PI plots below "A" line.

